TS1000

Robot Controller

INTERFACE MANUAL

Rev.3
April 1, 2020

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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The information contained in this manual is subject to change without prior notice to effect improvements.

Preface

This manual describes the type, function and handling method of external cables connecting the TS1000 robot controller with an external equipment.

This manual is intended for the system designers and manufacturing engineers.
The TS1000 robot controller can work in concert with the external equipment through digital input and output signals that can be programmed by the SCOL language. Also, system input signals that can allows external operation of the controller, system output signals informing an operator of the controller status (PLC processing function) and serial input and output signals that can be connected with the host computer, etc., are provided so that the user can easily construct an FA system.

- Instruction manuals which are referred to from this manual
  - Installation & Transport Manual (STE 71286)
  - Maintenance Manual (STE 71287)
  - Safety Manual (STE 71288)
  - User Parameter Manual (STE 71296)

! **CAUTION**	This manual does not contain any detailed descriptions on power and robot connection. For the connection of the power and robot, see the Installation & Transport Manual.

Cautions on Safety

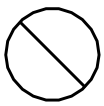
This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damages to assets and to assure correct use.
Make sure that you well understand the following details (indications and symbols) before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to damages due to destruction of assets or resources.

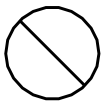
[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol.

! **CAUTION**	To perform the work ranging from robot installation to operation with safety, read through and through the Safety Manual provided separately before actually starting the work.

Maintenance and Inspection

To use the robot safety, strictly observe the following matters.

! DANGER	
 Prohibited	• NEVER burn, disassemble or charge the battery. Otherwise, it may explode.
 Mandatory	• Before performing the maintenance and inspection, be sure to turn off the main power switch of the controller. • When disposing of batteries, be sure to follow the user's regulations.

! CAUTION	
Disassembly Prohibited	• The user should NEVER replace or change parts other than those stipulated in the instruction manual. Otherwise, the performance will deteriorate, resulting in troubles.
 Mandatory	• To replace parts, use the spare parts designated by SHIBAURA MACHINE. • Carry out the maintenance and inspection on a regular basis. Otherwise, the equipment may go wrong or accidents will be caused.

! CAUTION	To perform the maintenance and inspection of the robot with safety, read through and through the Maintenance Manual provided separately before actually starting the work.

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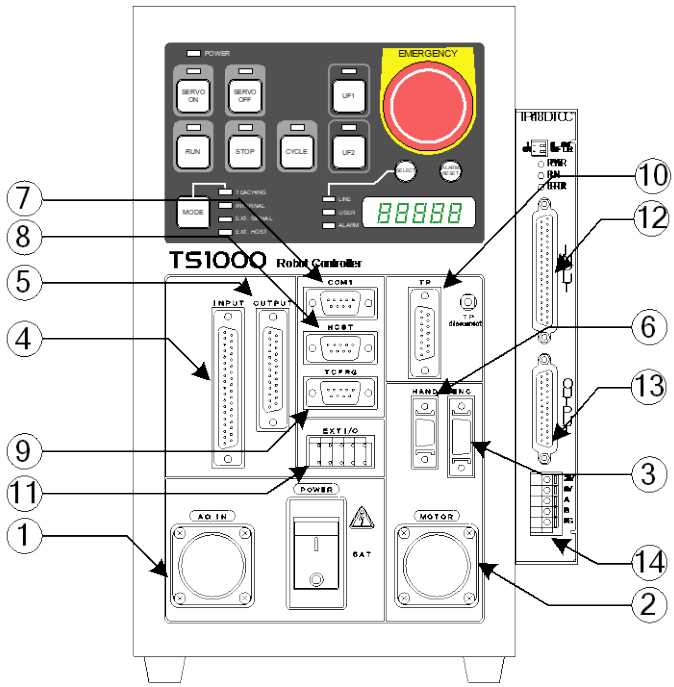
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Section 1 Type of External Cable

1.1 Layout and Name of Connectors

The TS1000 robot controller is connected with the robot and external equipment, using connectors provided on the front panel of the controller.



Front panel

Fig. 1.1 Layout and name of connectors

As shown in Fig. 1.1 above, all connectors are provided on the controller front panel. Thus, a space required for wiring can be provided on the front side alone, and no space for wiring is required on either the lateral side or rear side. (A space for ventilation is required on the lateral sides, however. If options are assembled, a space for mounting them is also necessary.)

1.2 TS1000 Power Cable "AC IN"

Fig. 1.1–① (with connector)

The power cable is an interface feeding the main power (2 phases, AC 170/250 V, 50/60 Hz) to the TS1000 robot controller. Connector "AC IN" is used.

For details, see the Installation & Transport Manual provided separately.

1.3 Robot Control Cable

1.3.1 Motor Drive Cable "MOTOR"

Fig. 1.1–② (with cable)
The motor drive cable connects the TS1000 robot controller and robot. It feeds the 3-phase AC power to each axis feed motor of the robot and connects the parking brake signal for securing the motor shaft. Connector "MOTOR" is used.
For details, see the Installation & Transport Manual provided separately.

1.3.2 Encoder Cable "ENC"

Fig. 1.1–③ (with cable)
The encoder cable is an interface which inputs the rotation angle detection encoder signal of each robot axis to the TS1000 robot controller. Connector "ENC" is used.
For details, see the Installation & Transport Manual provided separately.

1.4 External I/O Signal Cable

1.4.1 External Input Signal Cable "INPUT"

Fig. 1.1–④ (with dummy connector)
The external input signal cable is an interface recognizing each signal which is input to the TS1000 robot controller from the external equipment.
The external input signal comes in eight (8) digital input signals that can be programmed at the user's discretion, using the SCOL language or PLC function and thirteen (13) system input signals that allow operation of the controller from the external equipment. This signal, together with the external output signal in Para. 1.4.2, allows the robot controller to work in concert with the external equipment.
Also, three (3) system input signals of external servo ON, external servo OFF and external emergency stop are included in this cable. Connector "INPUT" is used.

1.4.2 External Output Signal Cable "OUTPUT"

Fig. 1.1–⑤ (with connector)
The external output signal cable is an interface which outputs each signal from the TS1000 robot controller to the external equipment.
The external output signal comes in eight (8) digital output signals that can be programmed at the user's discretion, using the SCOL language or PLC function and system output signals that output the controller status to the external equipment.
This signal, together with the external input signal in Para. 1.4.1, allows the robot controller to work in concert with the external equipment.
Connector "OUTPUT" is used.

1.5 Hand I/O Signal Cable "HAND"

Fig. 1.1–⑥ (Cable provided optionally; with robot side connector)
This signal is an interface which inputs and outputs the hand control signals for hand operation, etc. Among the robot control signal cables, the input/output signal cable which controls the robot hand end effector is called the "hand I/O signal cable" in this manual. These signals can be controlled by the robot language program. Also, they can be turned on and off manually from the teach pendant. Connector "HAND" is used.
For details, see the Installation & Transport Manual provided separately.
For the robot language, see the Robot Language Manual. Operations of hand input/output signals from the teach pendant are described in the Operator's Manual.

1.6 Serial I/O Signal Cable

1.6.1 Serial I/O Signal "COM1"

Fig. 1.1–⑦
Of the serial input and output signals of three (3) channels equipped on the TS1000 robot controller, the D-SUB 9-pin connector located on the upper center in Fig. 1.1 is COM1. COM1 is exclusively used for the RS232C and allows data communication with an image processing equipment or other FA equipment that can connect an RS232C interface. Connector "COM1" is used.

1.6.2 Serial I/O Signal "HOST"

Fig. 1.1–⑧

Of the serial input and output signals of three (3) channels equipped on the TS1000 robot controller, the D-SUB 9-pin connector located on the second from the upper center in Fig. 1.1 is HOST. HOST is exclusively used for the RS232C and allows transfer and saving of various parameters and updating of the system when connected with the host computer. Connector "HOST" is used.

1.6.3 TCPRGOS "TCPRG"

Fig. 1.1–⑨

The TS1000 robot controller uses exclusive sequence control tool "TCPRGOS" (option) to transfer, save and monitor sequence programs.
To connect with this sequence control tool, the D-SUB 9-pin connector located on the third from the upper center in Fig. 1.1 is TCPRG.
For the use of TCPRGOS, see the Simple PLC Function Manual (option) provided separately.

1.7 Teach Pendant Cable "TP"

Fig. 1.1–⑩ (with dummy plug)

This is an interface connecting the TS1000 robot controller and teach pendant (TP). The teach pendant is an option. By connecting the TP cable, creation of motion programs, manual robot guidance, etc., are possible through the teach pendant. Connector "TP" is used. The TP cable is secured to the teach pendant and cannot be disconnected from the teach pendant. The standard cable length is 5 m.

1.8 External I/O Cable "EXT I/O"

Fig. 1.1–⑪

This is an RS485 communication terminal connecting the TS1000 robot controller with the optional remote I/O module function (TR48DIOC module, etc.). Terminal block "EXT I/O" is used.
For details of the TR48DIOC, see the Extension I/O Function Manual (option).

1.9 Extension I/O Signal Cable

1.9.1 Extension Input Signal Cable "INPUT (TR)"

Fig. 1.1–⑫ (Attached to the option)
This is an interface which inputs an external signal from the external equipment to remote I/O module TR48DIOC. Like the external input signal cable, the extension input signal cable is provided with twenty-eight (28) digital input signals that can be programmed at the user's discretion, using the SCOL language or PLC function. The extension input signal, together with the external output signal or extension output signal, allows the robot controller to work in concert with the external equipment. Connector "INPUT (TR)" is used.

Note: To discriminate from the INPUT cable connector of the TS1000 robot controller, the extension input signal cable connector is referred to as "INPUT (TR)" in this manual.

1.9.2 Extension Output Signal Cable "OUTPUT (TR)"

Fig. 1.1–⑬ (Attached to the option)
This is an interface which outputs an external signal from remote I/O module TR48DIOC to the external equipment. Like the external output signal cable, the extension output signal cable is provided with twenty (20) digital output signals that can be programmed at the user's discretion, using the SCOL language or PLC function. The extension output signal, together with the external input signal or extension input signal, allows the robot controller to work in concert with the external equipment. Connector "OUTPUT (TR)" is used.

Note: To discriminate from the OUTPUT cable connector of the TS1000 robot controller, the extension output signal cable connector is referred to as "OUTPUT (TR)" in this manual.

1.9.3 Remote Signal Cable

Fig. 1.1–⑭ (Attached to the option)
This is an RS485 communication terminal which is connected with the EXT I/O cable in Para. 1.8 above.

Section 2 Power Cable

Connecting the power cable
To connect the TS1000 power cable, use the attached connector (JL04V–2E18–10PE–B; made by Japan Aviation Electronics Industry).

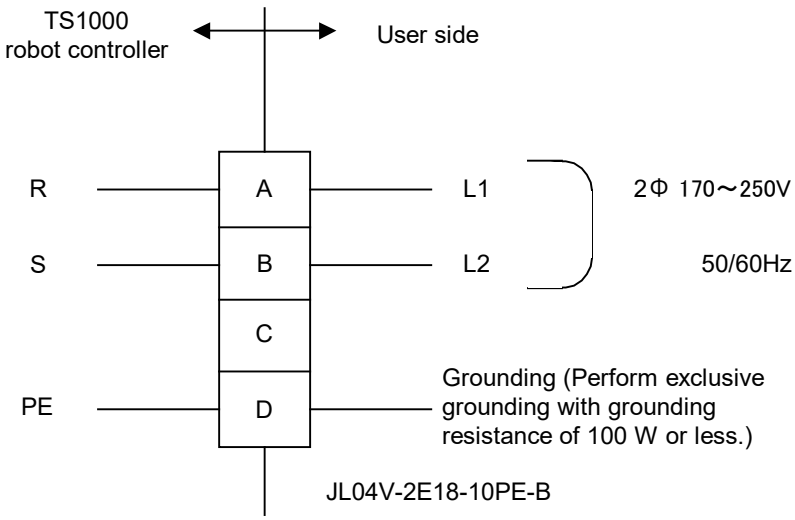


Fig. 2.1 Connection of power cable

For details of the power cable connection, see the Installation & Transport Manual provided separately.

Section 3 Robot Control Cable

3.1 TS1000 Motor Drive Cable

To connect the motors, use the attached cables.

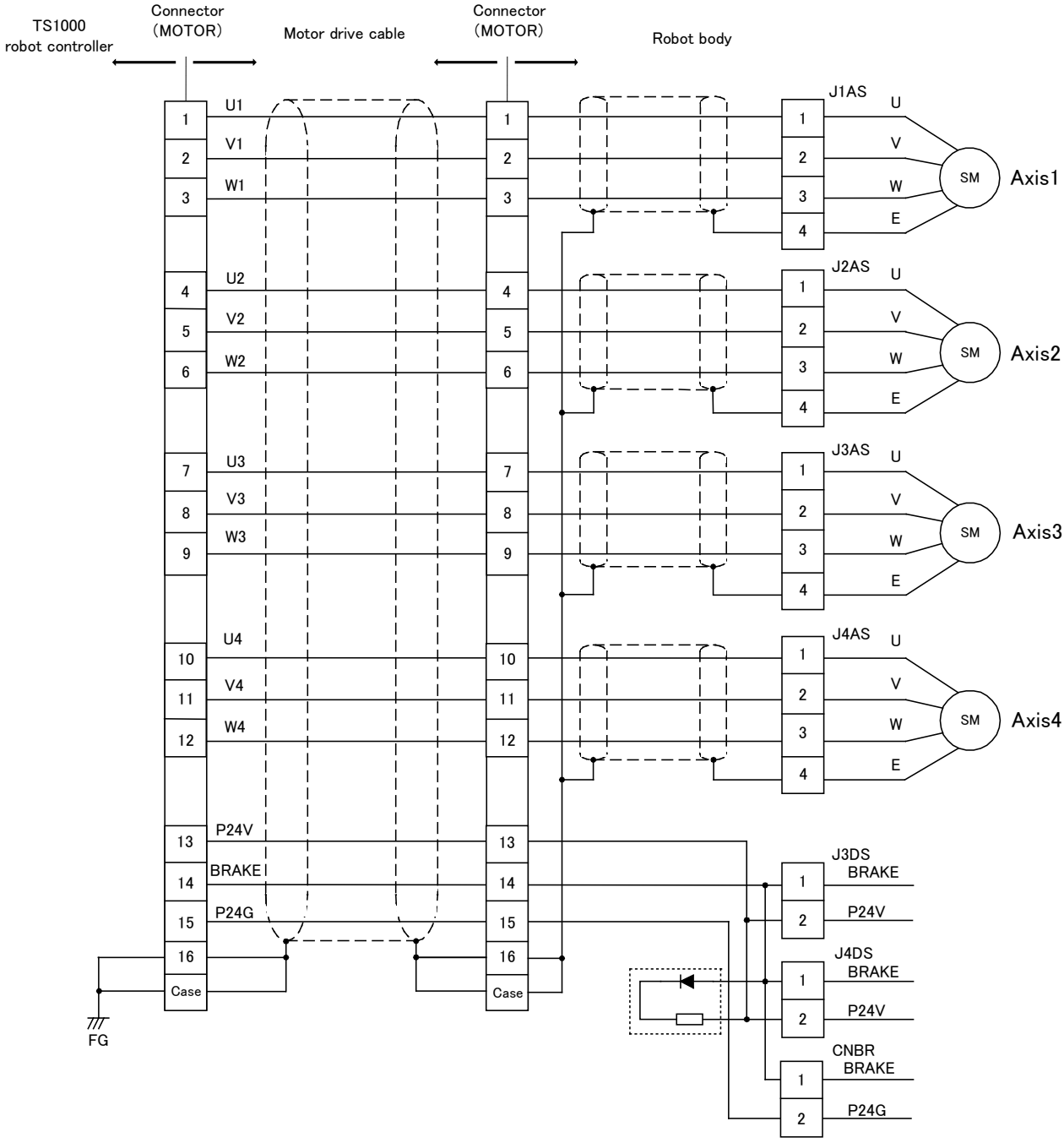


Fig. 3.1 Connection of motor drive cables

For the motor drive cable connection, see the Installation & Transport Manual.

3.2 TS1000 Encoder Cable

To connect the encoders, use the attached cables.

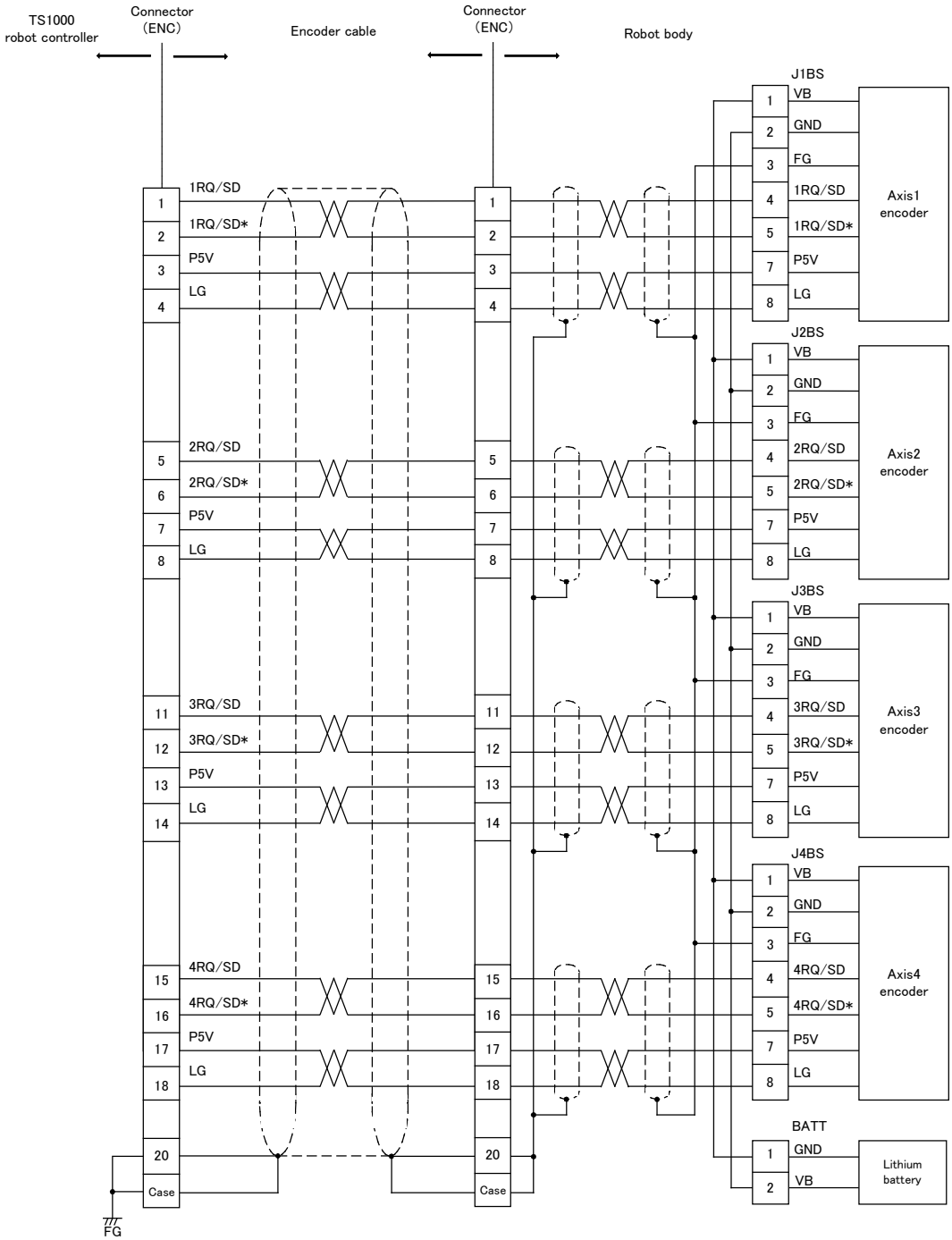


Fig. 3.2 Connection of encoder cables

For the encoder cable connection, see the Installation & Transport Manual.

Section 4 External I/O Signal Cable

4.1 External Input Signal Cable

4.1.1 Connection of External Input Signal Cable

To connect the external input signal cable, use the attached dummy connector [XM2A–3701 (pin type connector), XM2S–3711 (connector cover)]. Some system input signals (normal close contact) of this dummy connector are jumpered. Remove the jumpers, when necessary.
A cable with its single side soldered to the connector (TS1000 INPUT cable) is provided optionally. Contact our after-sale service agent in your territory.

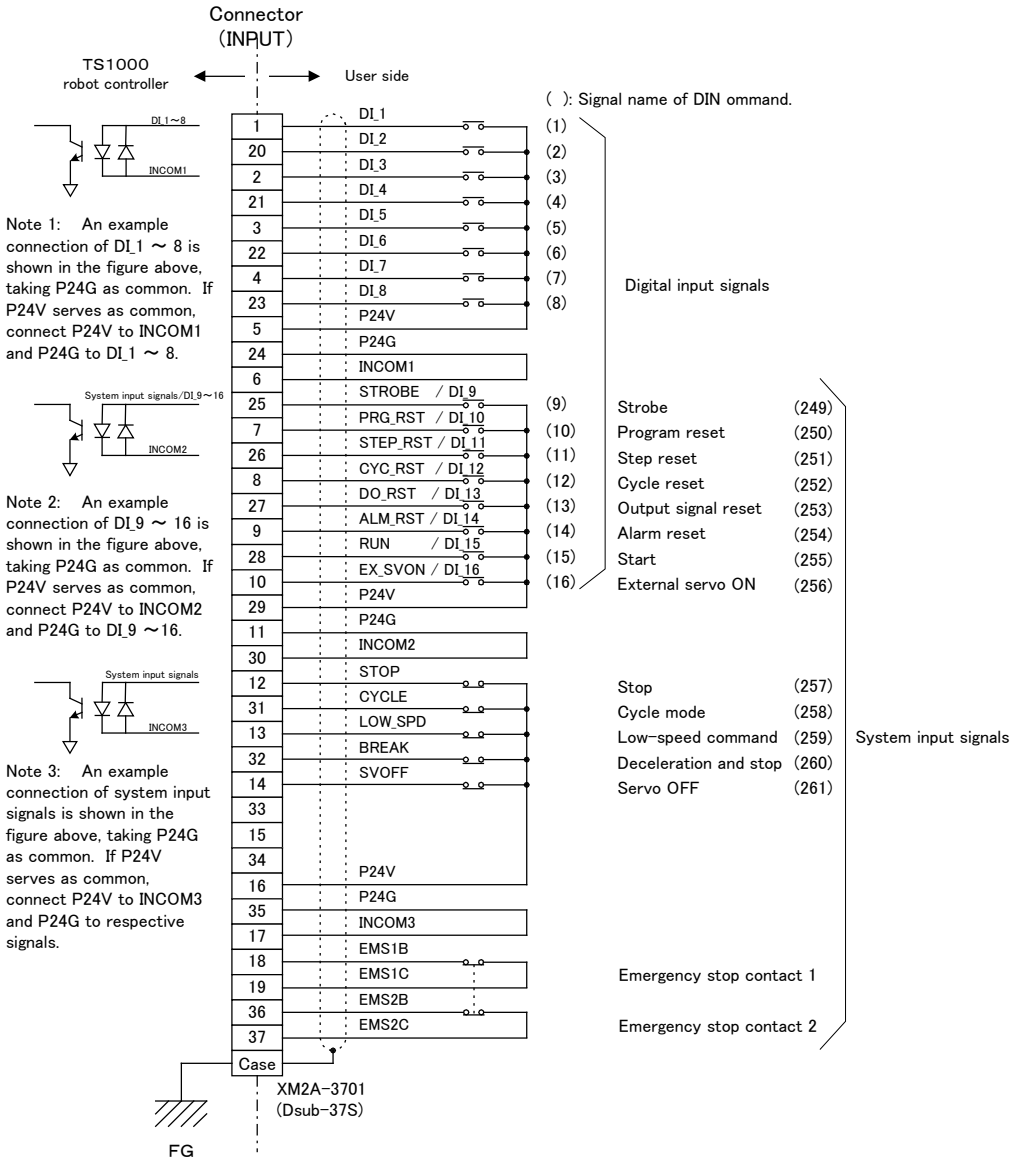
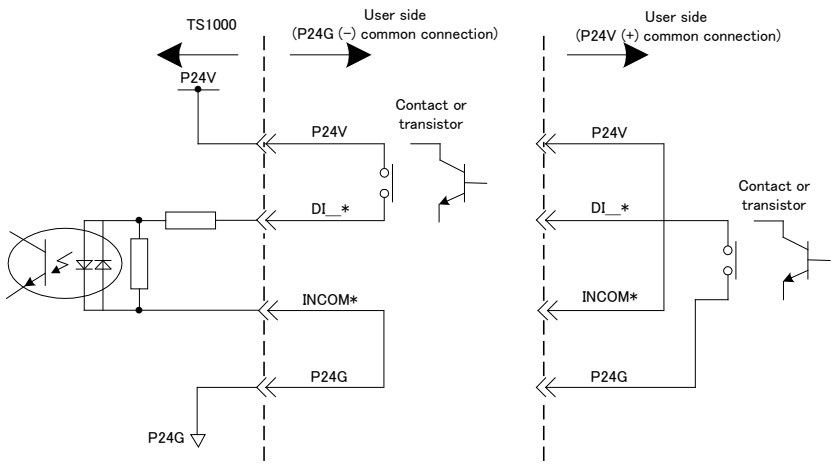
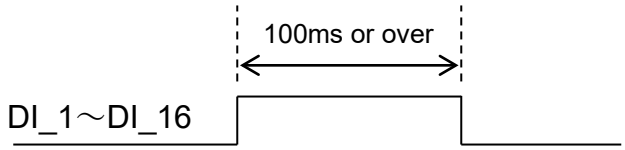


Fig. 4.1 Connection of external input signal cable

As shown in Fig. 4.1 above, the external input signals are non-voltage contact inputs or open collector inputs. Some of the system inputs (STOP, CYCLE, LOW_SPD, BREAK, SVOFF), emergency stop contact 1 and emergency stop contact 2 (two (2) contacts are interconnected) are normal close contact inputs, and the others are normal open contact inputs.
For the function, circuit to use, changeover of common wire, etc., see Para. 4.1.2.

4.1.2 Digital Input Signal

Designation	Digital input signal DI_1 ~ DI_8, DI_9 ~ DI_16 (system input signals)
Connector input terminal	Signals DI_1 ~ DI_8 are assigned to INPUT 1 ~ 4 pins and 20 ~ 23 pins. (See Fig. 4.1.) The system input signals assigned to INPUT 7 ~ 10 pins and 25 ~ 28 pins can be used as signals DI_9 ~ DI_16 by changing the user parameter. For details of the common wire connection, see the figure below (i.e., example of circuit).
Function	Each signal status of DI_1 ~ DI_16 can be identified by the robot program (DIN command) to branch the processing of program. Also, it is possible to perform interruptive processing of each signal (DI_1 ~ DI_16) with change in signal status monitored during the robot operation.
Input type	Non-voltage contact input or transistor open collector input.
Example of circuit (Input circuit structure)	
Signal logic	Input terminal Signal judgment Open OFF Short-circuit ON

Specifications of non-voltage contact and transistor	• Non-voltage contact specifications Contact rating DC24 V, 10 mA or over Circuit current: Approx. 7 mA Minimum current DC 24 V, 1 mA Contact impedance 100 Ω or less • Transistor specifications Withstand voltage between collector and emitter 30 V or over Current between collector and emitter 10 mA or over Circuit current: Approx. 7 mA Leakage current between collector and emitter 100 μA or less
Signal timing	When the pulse type input signals are used, the pulse width should be 100 ms or over. 

* When using the system input signals as the digital input signals:

User parameter [U13]
[U13] Select input signal (Default I/O mode only)
{Input 9–16} (0: System 1: User)
= 0 0 0 0 0 0 0 0

Specify "1" for a desired bit underlined above, and appropriate system input signal can serve as the digital input signal. Respective bits signify STROBE, PRG_RST, STEP_RST, CYC_RST, DO_RST, ALM_RST, RUN and EX_SVON when viewed from the left, which correspond to digital signals DI_9 to DI_16.

To use only system input signal STROBE as digital input signal DI_9, for instance, change the above bits as shown below.

1 0 0 0 0 0 0 0

To use all system input signals as the digital input signals, specify in the following manner.

1 1 1 1 1 1 1 1

After the above parameter has been changed, save the data, turn the power off and on again. Otherwise, the parameter will not be operative.

4.1.3 System Input Signal

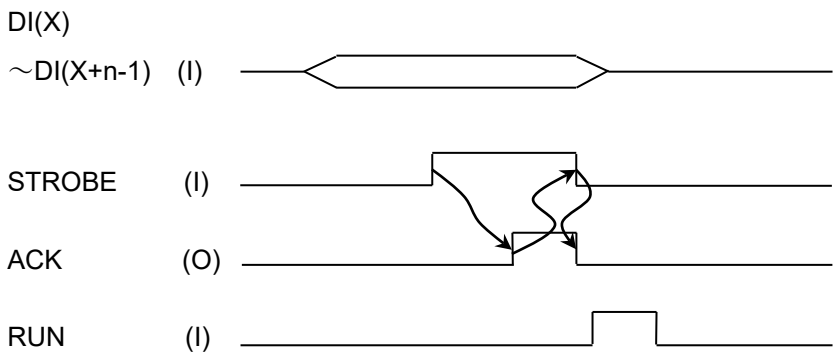
A total of thirteen (13) system input signals are available, which control STOP, CYCLE, etc., sent from the external equipment to the TS1000 robot controller.
The system input signal is provided with an exclusive input terminal for each function.
Of the thirteen (13) signals, eight (8) signals can be used as the digital input signals (DI_9 ~ DI_16) by changing the user parameter (initial setting).
The structure of the system input signal is quite the same as that of the digital input signal. See the descriptions in Para. 4.1.2. (However, the two (2) signals of emergency stop contact 1 and emergency stop contact 2 differ from the above specifications. See the descriptions on each signal.)

When inputting the system input signal, keep the input status until the output signal corresponding to each input is output to assure each signal input. The relationship between input signal and output signal is stipulated by the timing chart of each signal. Also, each system input signal becomes valid or invalid by means of the MODE (master mode) selector switch equipped on the control panel. Each signal ON mode is shown in Table 4.1 below.

Table 4.1 List of system input signal ON modes

Master mode / Designation	ON mode			
	TEACHING	INTERNAL	EXT. SIGNAL	EXT. HOST
STROBE (Strobe)			O	
PRG_RST (Program reset)			O	
STEP_RST (Step reset)			O	
CYC_RST (Cycle reset)			O	
DO_RST (Output signal reset)			O	
ALM_RST (Alarm reset)			O	
RUN (Start)			O	
EX_SVON (External servo ON)			O	
STOP (Stop)	O	O	O	O
CYCLE (Cycle operation mode)			O	
LOW_SPD (Low speed command)	O	O	O	O
BREAK (Deceleration and stop)	O	O	O	O
SVOFF (Servo OFF)	O	O	O	O
EMS*B ~ EMS*C (Emergency stop contacts 1 and 2)	O	O	O	O

O : ON mode

Designation	STROBE (Strobe)
Input terminal	INPUT 25 pin
signal name in DIN command	249
Signal logic	<table><tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr><tr><td>Open</td><td></td><td>O</td></tr><tr><td>Short-circuit</td><td>O</td><td></td></tr></table>
Function	Used to select an execution program for the TS1000 robot controller from the external equipment. The program number selected should use any successive "n" numbers (max. four (4) numbers) of external digital input signal, which are coded. 4 3 2 1 ← Program Number (max. four (4) bits) DI(X+n–1) --------- DI (X) ← External digital input signal * n = 1 ~ 4 For the program file name and register of it to the program number, and assignment of bits to external digital input signals, see the User Parameter Manual. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	 With the start of the STROBE signal, the above digital signals are read to select an appropriate program. After the program has been selected, the ACK signal turns on. Turn on the RUN signal and execute the program.

Cautions	The STROBE signal should not be input together with the PRG_RST, CYC_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid. If a file other than the current file is selected, the program is reset to step 1 and the values of variables are reset also.

* Program file name and register of it to program number, and assignment of bits to external digital input signals:

  To select an execution file (i.e., program selection), using digital input or extension input signals, assignment of bits to the controller input signals is necessary.

  User parameter [U07]
  [U07] Specify signal for EXTSELECT.
  {Signal No.} (1 –)
  {Bit length} (1 – 4)
  = 1 4

  Bits underlined = (Leading signal name in DIN command) (Bit length).
  "Signal name in DIN command" signifies an input signal number assigned in a program to run the internal sequence. (The signal name is predetermined in the program and cannot be changed by the user.)
  For the digital input signal, "signal name in DIN command" represents a number in parentheses as shown on the right side of the signal name in Fig. 4.1.

  (Leading signal name in DIN command):
  Specify the leading number of input signals to be used. (Allowable range: 1 ~ 16, 101 ~ 164)

  (Bit length):
  Specify the number of signals to be used. (Allowable range 1 ~ 4)

  Set value underlined (ex.) = 1 4
  This signifies that four (4) external digital input signals 1 ~ 4 are used.

Correspondence table
between [U07] set value (example) and [U08] program file name

Signal name in DIN command / Program file name ([U08] initial setting)	4	3	2	1
"PROG1"	0	0	0	0
"PROG2"	0	0	0	1
"PROG3"	0	0	1	0
"PROG4"	0	0	1	1
"PROG5"	0	1	0	0
"PROG6"	0	1	0	1
"PROG7"	0	1	1	0
"PROG8"	0	1	1	1
"PROG9"	1	0	0	0
"PROG10"	1	0	0	1
"PROG11"	1	0	1	0
"PROG12"	1	0	1	1
"PROG13"	1	1	0	0
"PROG14"	1	1	0	1
"PROG15"	1	1	1	0
"PROG16"	1	1	1	1

When bits underlined = 1 2, two (2) external digital input signals 1 and 2 are used, and the number of program files selected is four (4) from "PROG1" ~ "PROG4".

To register the program file name to the program number, use user parameter [U08] (see below).

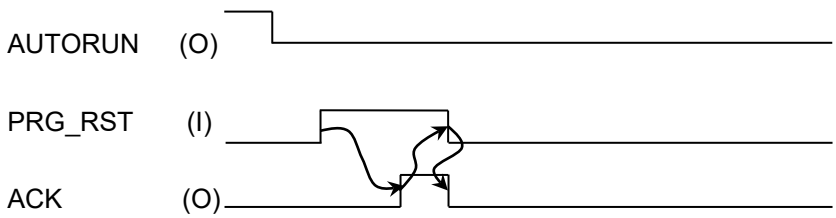
[U08] EXTSELECT file
= "PROG1"
= "PROG2"
= "PROG3"
= "PROG4"
= "PROG5"
= "PROG6"
= "PROG7"
= "PROG8"
= "PROG9"
= "PROG10"
= "PROG11"
= "PROG12"
= "PROG13"
= "PROG14"
= "PROG15"
= "PROG16"

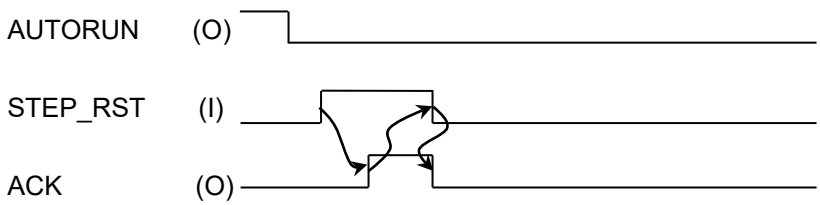
The initial setting is as shown above. Specify a file name you registered beforehand for the underlined of "PROG**".

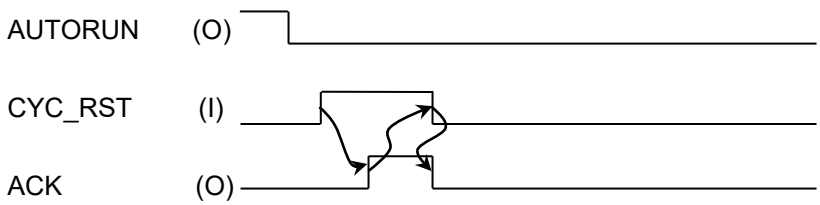
Example:
[U08] EXTSELECT file
= "AAA"
= "BBB"
= "CCC"
= "DDD"

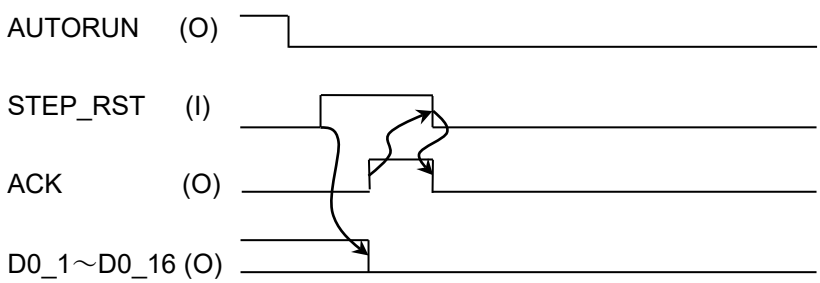
Specify the above program names in advance. (If the following steps are taken without registering the program names, a "Compile Error" occurs.) Then specify zero (0) for both bits 1 and 2 of DIN command, which are set under [U07] (assume that [U07] = 1 2) and input the STROBE signal. Program "AAA" is automatically selected now.

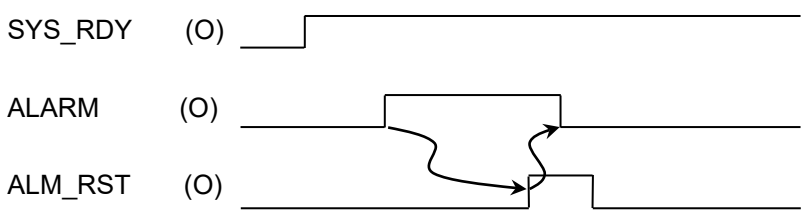
After the above parameter has been changed, save the data, turn the power off and on again. Otherwise, the parameter will not be operative.

<table>
<tr><td>Designation</td><td>PRG_RST (Program reset)</td></tr>
<tr><td>Input terminal</td><td>INPUT 7 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>250</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to reset a currently stopped program to step 1. The value of each variable is also reset to zero (0).
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. The PRG_RST signal should not be input together with the STROBE, CYC_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.
2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.</td></tr>
</table>

<table>
<tr><td>Designation</td><td>STEP_RST (Step reset)</td></tr>
<tr><td>Input terminal</td><td>INPUT 26 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>251</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to reset a currently stopped program to step 1. The value of each variable used in the program remains unchanged.
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. The STEP_RST signal should not be input together with the STROBE, PRG_RST, CYC_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.
2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.</td></tr>
</table>

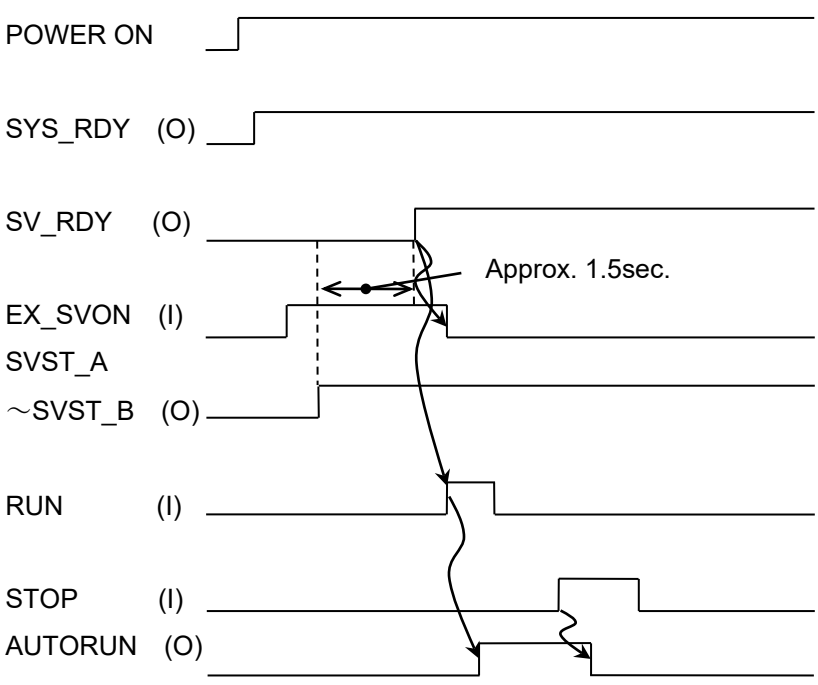
<table>
<tr><td>Designation</td><td>CYC_RST (Cycle reset)</td></tr>
<tr><td>Input terminal</td><td>INPUT 8 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>252</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to reset a currently stopped program to the step labeled "RCYCLE". The value of each variable used in the program remains unchanged.
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. The CYC_RST signal should not be input together with the STROBE, PRG_RST, STEP_RST or DO_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid.
2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.</td></tr>
</table>

Designation	DO_RST (Output signal reset)
Input terminal	INPUT 27 pin
signal name in DIN command	253
Signal logic	<table><tr><th>Signal judgment / Signal terminal</th><th>ON</th><th>OFF</th></tr><tr><td>Open</td><td></td><td>O</td></tr><tr><td>Short-circuit</td><td>O</td><td></td></tr></table>
Function	Used to reset digital output signals (DO_1 ~ DO_16, DO_101 ~ DO_120, DO_133 ~ DO_152) of the TS1000 robot controller from the external equipment. When reset, all signals of DO_1 ~ DO_16 turn off. When DO_8 ~ DO_16 are set as the system output signals, however, they are turned on. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	
Cautions	1. The DO_RST signal should not be input together with the STROBE, PRG_RST, CYC_RST or STEP_RST signal. Because the ACK signal is used in common, only the first signal which is input becomes valid and all other signals become invalid. Neither system output signals nor hand output signals are reset. 2. This signal cannot be used while AUTORUN (automatic operation mode ON) is set ON.

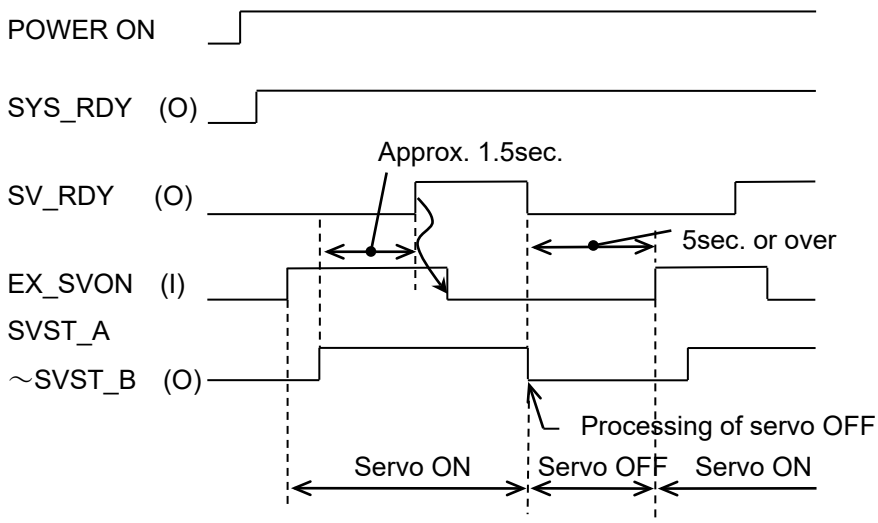
<table>
<tr><td>Designation</td><td>ALM_RST (Alarm reset)</td></tr>
<tr><td>Input terminal</td><td>INPUT 9 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>254</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to cancel an alarm from the external equipment, which occurred while the robot controller was able to start.
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>If an alarm of the emergency stop level which will not allow the processing of EX_SVON, or EMSST_A ~ EMSST_B is output, alarm reset by the ALM_RST signal is not possible.</td></tr>
</table>

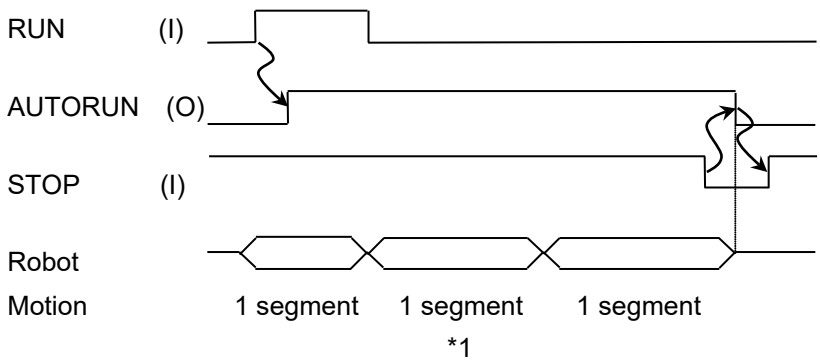
Designation	RUN (Start)
Input terminal	INPUT 28 pin
signal name in DIN command	255
Signal logic	

Signal judgment / Signal terminal	ON	OFF
Open		O
Short-circuit	O	

Function	Used to start a program registered in the TS1000 robot controller from the external equipment to execute an automatic cycle operation. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	

Cautions	After servo ON, automatic operation starts with the start of the RUN signal. It takes about one (1) second from the input of EX_SVON to the time when the robot is actually ready to work. Thus, it is necessary to program a delay of 1.5 seconds or over between the SVST_A ~ SVST_B signal ON and the RUN signal ON. Set ON the RUN signal only after the SV_RDY signal is ON. If the RUN signal is input without a delay, it is neglected and the automatic operation may not be started.

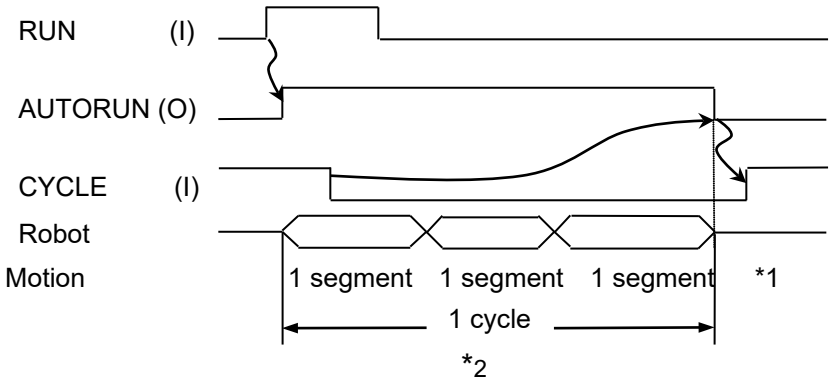
<table>
<tr><td>Designation</td><td>EX_SVON (External input servo ON)</td></tr>
<tr><td>Input terminal</td><td>INPUT 10 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>256</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td></td><td>O</td></tr>
<tr><td>Short-circuit</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to turn on the servo driver main power from the external equipment. Once the servo power is turned on, it is maintained even after this signal turns off.
This signal can be used only in the EXT. SIGNAL mode.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. It takes about one (1) second from the servo ON to the time when the robot is actually ready to work. Therefore, program so that the RUN signal, etc., can turn on only after the SV_RDY signal turns on.
2. To prevent an internal damage, the servo cannot be turned on about 4.5 seconds after it is turned off. To turn the servo on again, wait at least five (5) seconds after the SVST_A ~ SVST_B signal turns off.</td></tr>
</table>

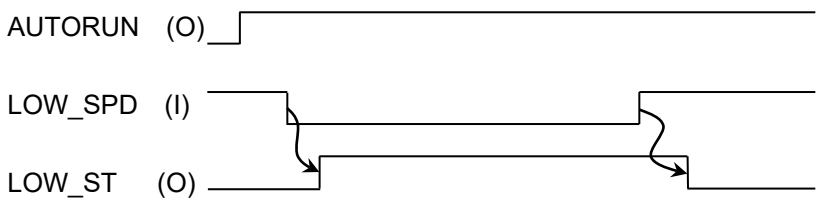
<table>
<tr><td>Designation</td><td>STOP (Stop)</td></tr>
<tr><td>Input terminal</td><td>INPUT 12 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>257</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to stop executing a program registered in the TS1000 robot controller from the external equipment. When this signal is open, the program stops after the current motion command has been executed.
When this signal is open, the robot cannot be operated.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 

*1 Duration from the start of one motion command to just before the start of next motion command is called "1 segment".</td></tr>
<tr><td>Cautions</td><td>1. When the RUN command is executed after cancel of STOP, the program restarts from the step next to the interrupted step.
2. RUN signal input is ineffective at the input of STOP signal.
3. Unless this signal is used, short-circuit INPUT 12 according to the common wire connection.</td></tr>
</table>

Designation	CYCLE (Cycle operation mode)
Input terminal	INPUT 31 pin
signal name in DIN command	258
Signal logic	

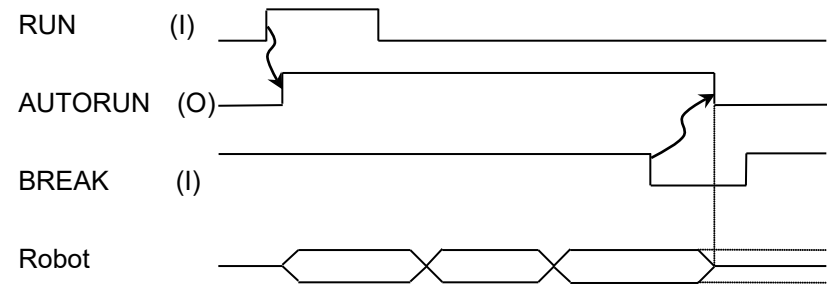
Signal terminal \ Signal judgment	ON	OFF
Open	O	
Short-circuit		O

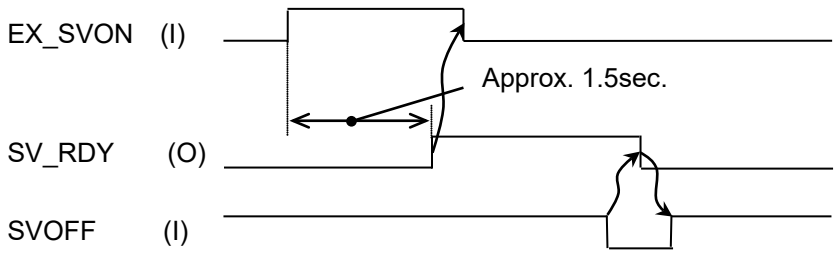
Function	Used to stop from the external equipment a program registered in the TS1000 robot controller after current one (1) cycle operation has been executed during automatic operation. This signal can be used only in the EXT. SIGNAL mode.
Signal timing	 *1 Duration from the start of one motion command to just before the start of next motion command is called "1 segment". *2 Duration from the top of the main program to the END command is called "1 cycle".
Cautions	1. When the RUN command is executed after cancel of CYCLE, continuous operation starts from the next cycle of the stopped program. 2. To make this signal operative, keep the signal valid time until the operation stops. If this signal becomes invalid before the operation stops, the automatic operation continues. 3. Unless this signal is used, short-circuit INPUT 31 as necessary according to the common wire connection.

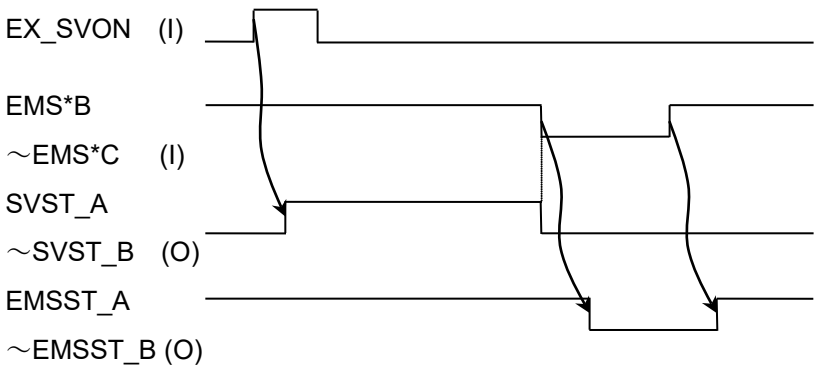
<table>
<tr><td>Designation</td><td>LOW_SPD (Low speed command)</td></tr>
<tr><td>Input terminal</td><td>INPUT 13 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>259</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to cause the robot operation speed to low speed from the external equipment.
The robot operates at a low speed (the low speed command is valid) while this signal is open.
The robot operation speed in the low speed mode can be set by means of the parameter. (Initial set value: 25 %)
When this signal is short-circuited, the previously set value (initial set value: 100 %) takes effect again.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. If the speed override lower than the parameter set value is used, even if the LOW_SPD signal is made valid, the speed override value remains unchanged.
2. Unless this signal is used, short-circuit INPUT 13 according to the common wire connection.</td></tr>
</table>

Designation	BREAK (Deceleration and stop)
Input terminal	INPUT 32 pin
signal name in DIN command	260
Signal logic	

Signal terminal \ Signal judgment	ON	OFF
Open	O	
Short-circuit		O

Function	Used to stop the robot motion from the external equipment. The robot slows down and stops at the same time that this signal is open. After the stop, the robot enters a STOP (RETRY) status. Even if this signal is short-circuited again after the stop of robot motion, the robot will not operate. To restart the robot, short-circuit this signal, then execute the RUN command. If this signal is open, the robot cannot be started. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	 *1 Duration from the start of one motion command to just before the start of next motion command is called "1 segment".
Cautions	1. If the robot is operating, processing of execution is interrupted and the robot slows down and stops. 2. Unless this signal is used, short-circuit INPUT 32 according to the common wire connection.

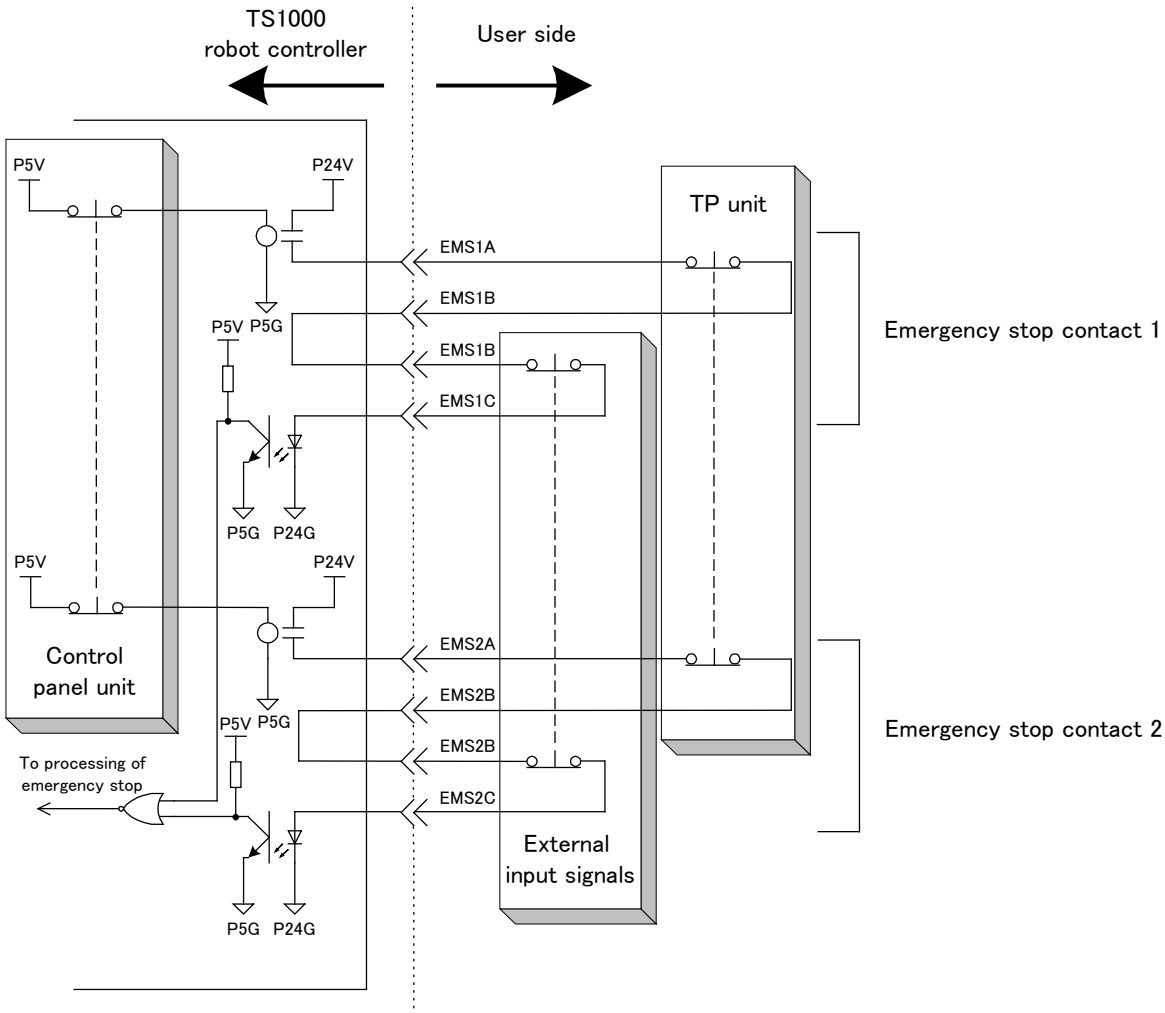
<table>
<tr><td>Designation</td><td>SVOFF (Servo OFF)</td></tr>
<tr><td>Input terminal</td><td>INPUT 14 pin</td></tr>
<tr><td>signal name
in DIN command</td><td>261</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Signal terminal</td><td>ON
(Servo OFF)</td><td>OFF
(Normal)</td></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to turn off the servo driver main power from the external equipment.
While this signal is open, the servo power is turned off.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. While this signal is open, the servo power cannot be turned on in any mode.
2. Unless this signal is used, short-circuit INPUT 14 according to the common wire connection.</td></tr>
</table>

<table>
<tr><td>Designation</td><td>EMS*B ~ EMS*C (Emergency stop contacts 1 & 2)</td></tr>
<tr><td>Input terminal</td><td>Between INPUT 18 ~ INPUT 19 (Emergency stop contact 1)
Between INPUT 36 ~ INPUT 37 (Emergency stop contact 2)</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><th>Signal judgment / Signal terminal</th><th>ON (Emergency stop)</th><th>OFF (Normal)</th></tr>
<tr><td>Open</td><td>O</td><td></td></tr>
<tr><td>Short-circuit</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>Used to emergency-stop the robot from the external equipment.
While this signal is open, the processing of robot emergency stop is executed.
Use this signal by connecting a safety device such as external emergency stop switch, photoelectric type sensing safety device and safety mat switch.
When the emergency stop contact is open, system output signals EMSST_A ~ EMSST_B are short-circuited.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
</table>

Cautions	1. While this signal is open, the servo power cannot be turned on in any mode. 2. EMS*A ~ EMS*B are assumed to have two (2) normal close contacts interconnected, which should be turned on and off at the same time. If there is a delay in contact operation, the non-conformity detecting function of hardware works to effect an emergency stop. When this happens, the system can be restored only by tuning off and on again the controller power. For the contact structure of the emergency stop switch, see the descriptions on the emergency stop signal line given below. 3. Unless this signal is used, short-circuit INPUT 18 ~ INPUT 19, and INPUT 36 ~ INPUT 37.

* Emergency stop signal line

The connection diagram of the emergency stop switch is shown below.



4.1.4 Jumper of Safety Measure Signal

Of the system input signal cables, the following signals are used for the safety measures.

System input signal cables ····· INPUT 12 (STOP)
INPUT 14 (SVOFF)
INPUT 32 (BREAK)
INPUT 18, 19 (EMS1B ~ EMS1C)
INPUT 36, 37 (EMS2B ~ EMS2C)
INPUT 17, 35 (INCOM3 ~ P24G)

For the dummy connector provided with the TS1000 robot controller, these signals are already jumpered. If these signals are used or changed, perform wiring with the jumper of connector removed. When operating the robot without using system input signals, be sure to connect the attached dummy connector to the INPUT connector on the controller side.
Unless the following signals are used as the external signals, jumper the cables also.

INPUT 13 (LOW_SPD)
INPUT 31 (CYCLE)

Jumper of connector [P24G(–) common is assumed as per Fig. 4.1.]

INPUT			
12–16	14–16	32–16	36–37
(13–16)	(31–16)	18–19	17–35

! **CAUTION**	Unless the signals of SVOFF and emergency stop contacts 1 and 2 are jumpered, the controller servo power cannot be turned on.

4.2 External Output Signal Cable

4.2.1 Connection of External Output Signal Cable

To connect the external output signal cable, use the attached connector [XM2A–2501 (pin type connector), XM2S–2511 (connector cover)].
A cable with its single side soldered to the connector (TS1000 OUTNPUT cable) is provided optionally. Contact our after-sale service agent in your territory.

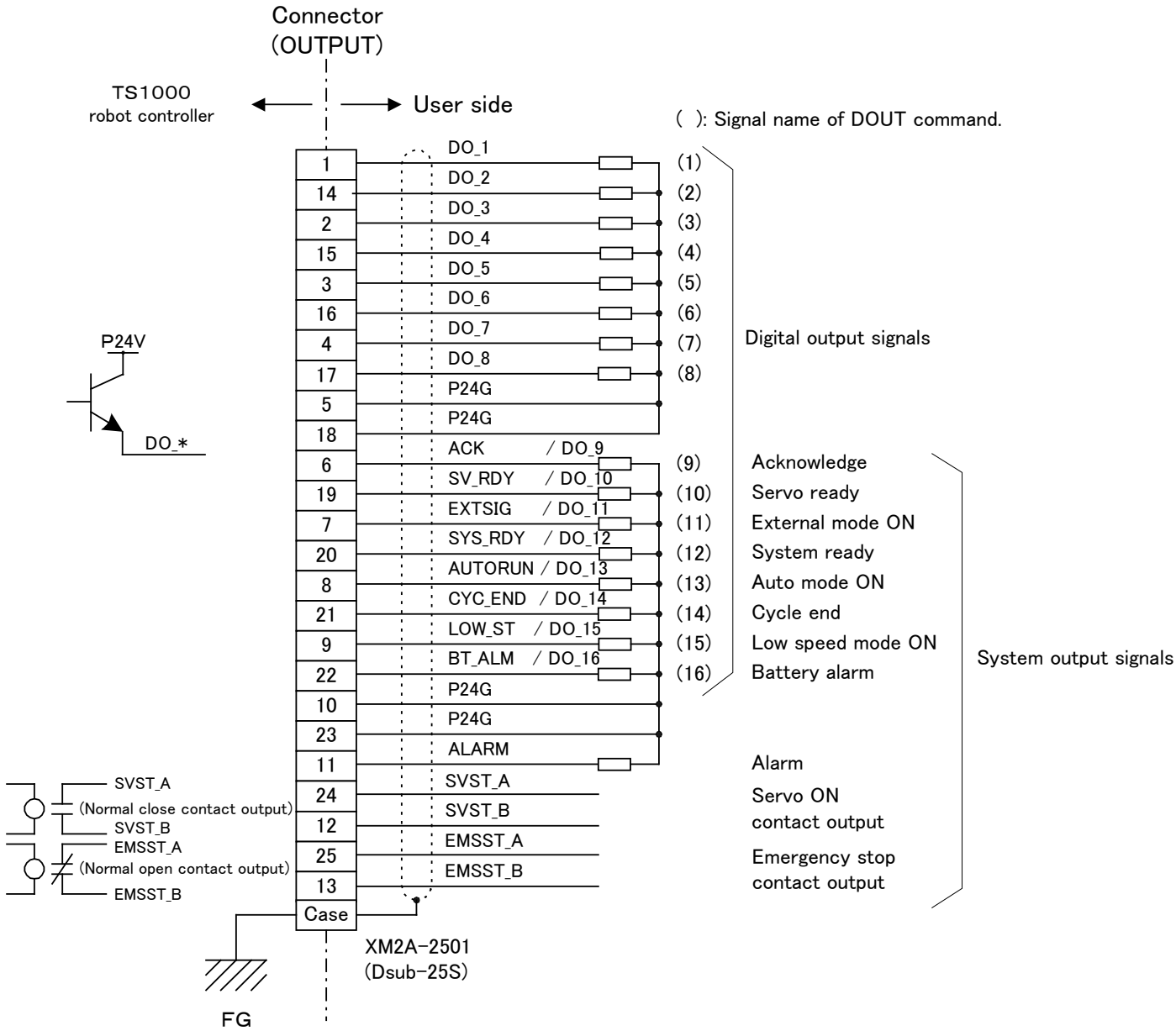
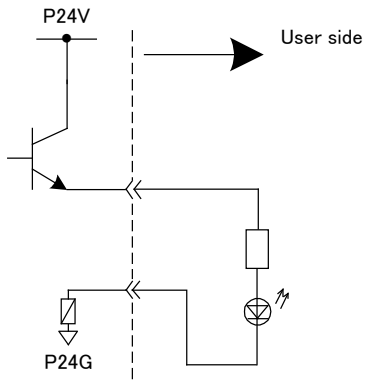
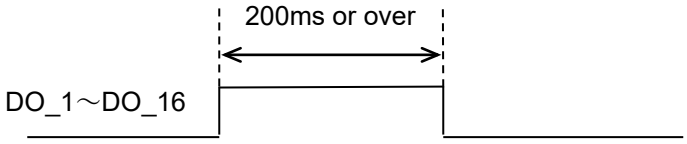
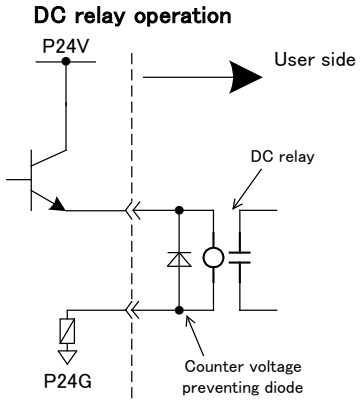


Fig. 4.2 Connection of external output signal cable

As shown in Fig. 4.2 above, all sixteen (16) digital output signals are transistor outputs. Two (2) system output signals (i.e., servo ON contact output and emergency stop contact output) are non-voltage relay contact outputs, and all other system output signals are transistor outputs.
For the function, circuit to use, etc., see Para. 4.2.2.

4.2.2 Digital Output Signal

Designation	Digital output signal DO_1 ~ DO_8, DO_9 ~ DO_16 (system output signals)
Connector output terminal	Signals DO_1 ~ DO_8 are assigned to OUTPUT 1 ~ 4 pins and 14 ~ 17 pins. (See Fig. 4.2.) The system output signals assigned to OUTPUT 6 ~ 9 pins and 19 ~ 22 pins can be used as signals DO_9 ~ DO_16 by changing the user parameter.
Function	ON/OFF of signals DO_1 ~ DO_16 and pulse output can be performed by the robot program (BCDOUT command and PULOUT command).
Output type	Transistor output
Output circuit structure	
Electric rating	Rated voltage: DC24 V Rated current: 100 mA (max.) (Maximum allowable current per 1 common is 0.4 A.) **Caution:** If the current which exceeds the rated output current is supplied, the output device may be damaged or the printed board may be burnt. To avoid this, be sure to use within the rated output current.

Signal timing	When performing pulse output by the PULOUT command, the output pulse width should be 200 ms or over. 
Example of circuit	

* When using the system output signals as the digital output signals:

User parameter [U14]
[U14] Select output signal (Default I/O mode only)
{Output 9–16} (0: System 1: User)
= 0 0 0 0 0 0 0 0

Specify "1" for a desired bit underlined above, and appropriate system output signal can serve as the digital output signal. Respective bits signify ACK, SV_RDY, EXTSIG, READY, AUTORUN, CYC_END, LOW_ST and BT_ALM when viewed from the left, which correspond to digital signals DO_9 to DO_16.

To use only system output signal ACK as digital output signal DO_9, for instance, change the above bits as shown below.

1 0 0 0 0 0 0 0

To use all system output signals as the digital output signals, specify in the following manner.

1 1 1 1 1 1 1 1

After the above parameter has been changed, save the data, turn the power off and on again. Otherwise, the parameter will not be operative.

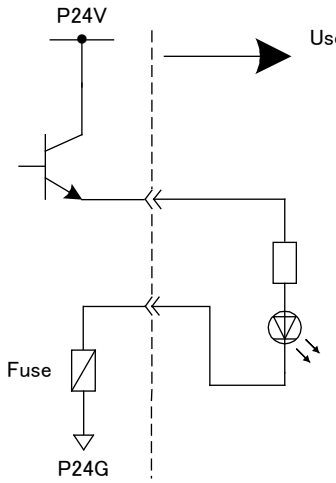
4.2.3 System Output Signal

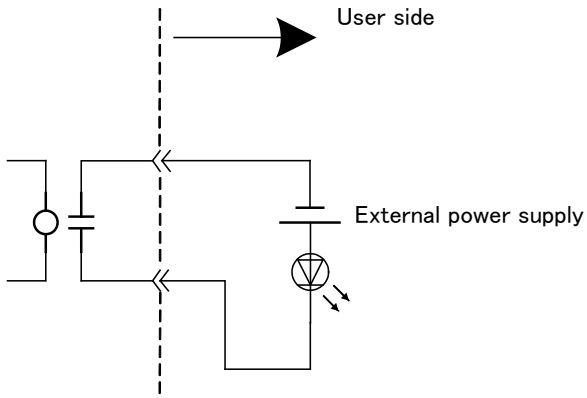
A total of eleven (11) system output signals are available. Of them, nine (9) signals are used to output the run status of the TS1000 robot controller and the remaining two (2) signals are output via relay contact.
The system output signal is provided with an exclusive output terminal for each function. Of the nine (9) system output signals, eight (8) signals can be used as the digital output signals by changing the user parameter (initial setting).
The two (2) relay output signals are SVST_A ~ SVST_B and EMSST_A ~ EMSST_B. The former signal (SVST_A ~ SVST_B) is of a normal open contact type and the latter signal (EMSST_A ~ EMSST_B) is of a normal close contact type. They differ in contact operation at the time of output signal ON/OFF. When using, take careful precautions not to mistake.
The specifications of the system output signals are as follows:

- Output type
  (1) Transistor output (system output signal)
  (2) Relay contact output (SVST_A ~ SVST_B, EMSST_A ~ EMSST_B)

  Note: SVST_A ~ SVST_B : Normal open contact output
  EMSST_A ~ EMSST_B : Normal close contact output

- Output circuit structure
  (1) Transistor output (2) Relay contact output



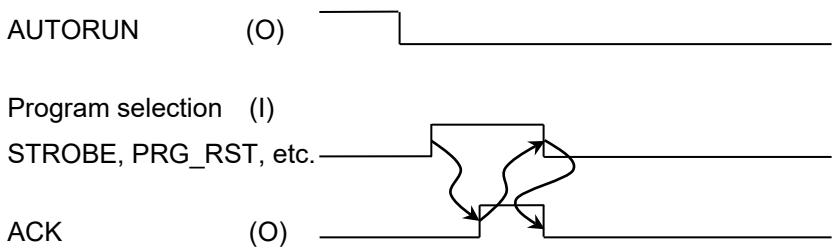


* The above figure exemplifies a normal open contact output structure. The servo ON contact output is of a normal open contact type and the emergency stop contact output is of a normal close contact type. Take utmost care when using.

- Transistor specification
  Rated voltage: DC 24 V
  Rated current: 100 mA max.

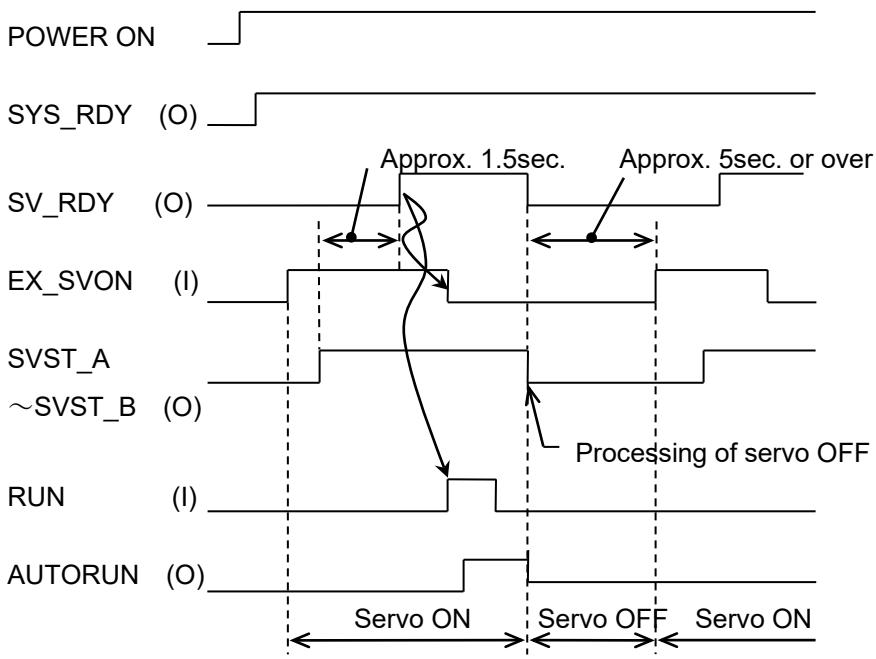
- Relay contact specification
  Maximum rating: AC125 V, 0.5 A / DC60 V, 1.0 A

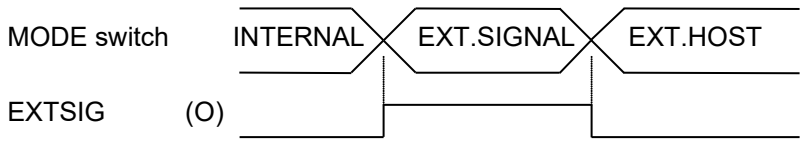
! **CAUTION**	If the current which exceeds the rated output current is supplied, the output device may be damaged or the printed board may be burnt. To avoid this, be sure to use within the rated output current.

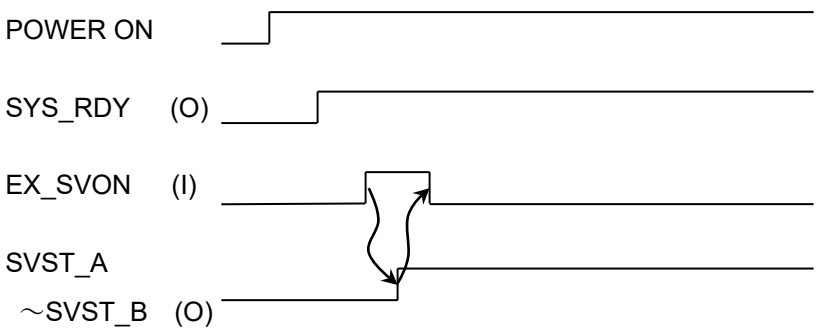
<table>
<tr><td>Designation</td><td>ACK (Acknowledge)</td></tr>
<tr><td>Output terminal</td><td>OUTPUT 6 pin</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment
Output terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>High</td><td>O</td><td></td></tr>
<tr><td>Low</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>This is a response signal to the STROBE, PRG_RST, STEP_RST, CYC_RST and DO_RST signals. When one of these signals is input, the ACK signal is sent back to inform that appropriate processing has finished.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>If two (2) or more input signals shown above are input at the same time, only the signal which was input first is processed, then the ACK signal is output.</td></tr>
</table>

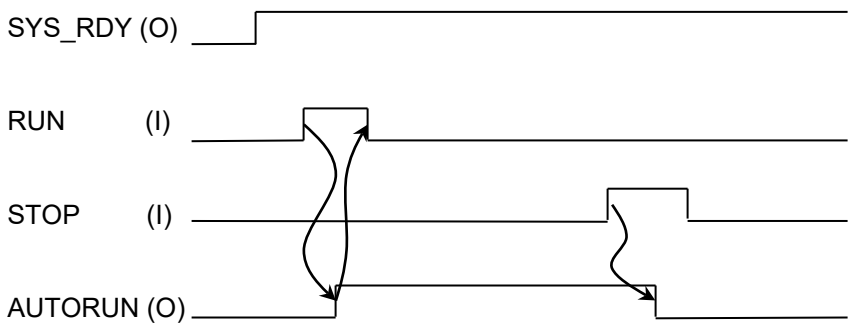
Designation	SV_RDY (Servo ready)
Output terminal	OUTPUT 19 pin
Signal logic	

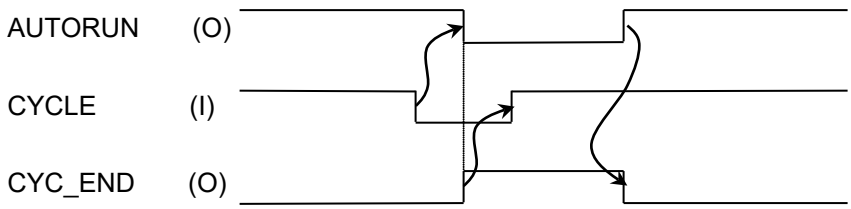
Signal judgment / Output terminal	ON	OFF
High	O	
Low		O

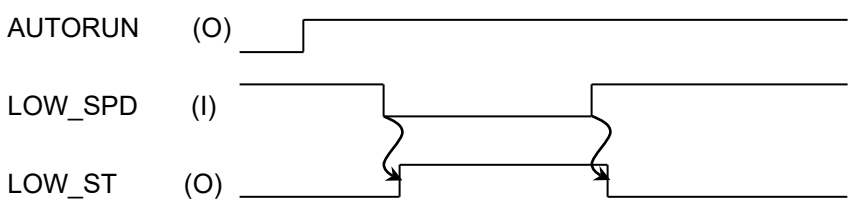
Function	This signal indicates that the servo power of the TS1000 robot controller can be turned on. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	1. Program so that the RUN signal can be turned on only after the SV_RDY signal turns on. 2. To prevent an internal damage, the servo cannot be turned on about 4.5 seconds after it is turned off. To turn the servo on again, wait at least five (5) seconds after the SVST_A ~ SVST_B signal turns off.

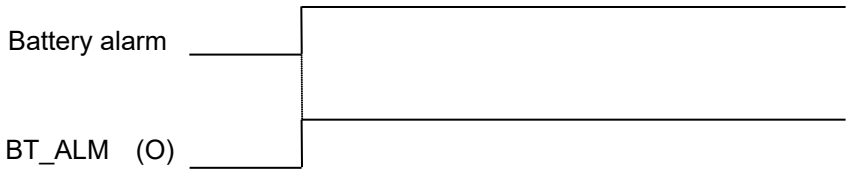
<table>
<tr><td>Designation</td><td>EXTSIG (External mode ON)</td></tr>
<tr><td>Output terminal</td><td>OUTPUT 7 pin</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Output terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>High</td><td>O</td><td></td></tr>
<tr><td>Low</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>This signal turns on when the MODE switch of the TS1000 robot controller is set to "EXT. SIGNAL".</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>When the EXT. SIGNAL mode is selected by means of the MODE switch, all system input signals become operative.</td></tr>
</table>

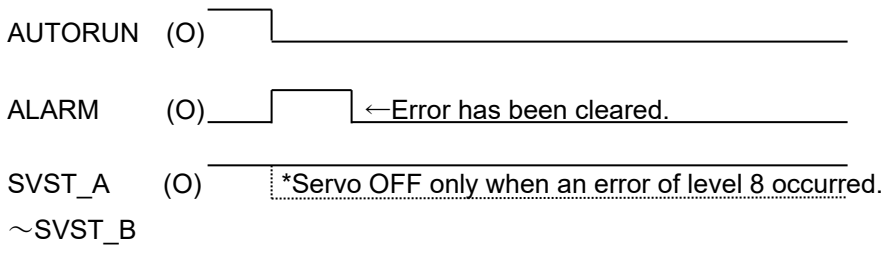
Designation	SYS_RDY (System ready)
Output terminal	OUTPUT 20 pin
Signal logic	<table><tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on when the robot can start after the controller main power is turned on. With this signal, it is possible to confirm that the robot can be operated. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	

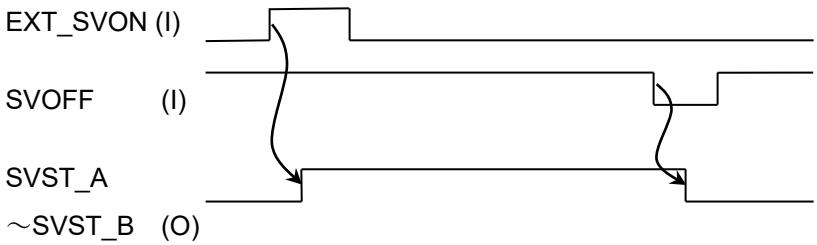
<table>
<tr><td>Designation</td><td>AUTORUN (Auto mode ON)</td></tr>
<tr><td>Output terminal</td><td>OUTPUT 8 pin</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Output terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>High</td><td>O</td><td></td></tr>
<tr><td>Low</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>This signal turns on when the robot is operating in the automatic operation mode.
When the RUN signal is made valid after the SYS_RDY signal is turned on, automatic operation starts with this signal output.
This signal remains on as long as the robot is operating in the automatic operation mode.
This signal turns on when the INTERNAL, EXT. SIGNAL or EXT. HOST mode is selected by means of the MODE switch of the TS1000 robot controller.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>This signal will not turn on while the TEACHING mode is selected by means of the MODE switch of the TS1000 robot controller.</td></tr>
</table>

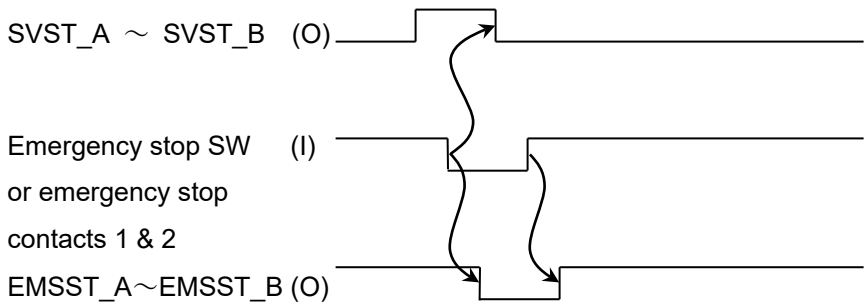
<table>
<tr><td>Designation</td><td>CYC_END (Cycle end)</td></tr>
<tr><td>Output terminal</td><td>OUTPUT 21 pin</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><th>Signal judgment / Output terminal</th><th>ON</th><th>OFF</th></tr>
<tr><td>High</td><td>O</td><td></td></tr>
<tr><td>Low</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>This signal turns on after the stop of 1-cycle automatic operation only when the CYCLE signal is made valid in the program execution (RUN) mode and an automatic operation is executed.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>1. This signal turns off when the RUN mode takes effect.
2. This signal turns on also when the cycle operation has been stopped by the STOP, BREAK or ALARM signal.</td></tr>
</table>

Designation	LOW_ST (Low speed mode ON)
Output terminal	OUTPUT 9 pin
Signal logic	<table><tr><td>Signal judgment / Output terminal</td><td>ON</td><td>OFF</td></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on while the robot is operating in the low speed mode by the input of system input signal LOW_SPD. This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	

Designation	BT_ALM (Battery alarm)
Output terminal	OUTPUT 22 pin
Signal logic	<table><tr><td>Signal judgment / Output terminal</td><td>ON</td><td>OFF</td></tr><tr><td>High</td><td>O</td><td></td></tr><tr><td>Low</td><td></td><td>O</td></tr></table>
Function	This signal turns on if a battery alarm has occurred in the robot or robot controller. The battery alarm detects all encoders of axis 1 to axis 5 and battery level in the main control board (X8LC). This signal can always be used, irrespective of the master mode selected by means of the MODE switch.
Signal timing	
Cautions	If the battery alarm has occurred, replace the battery immediately by identifying a battery whose voltage has dropped and referring to the Maintenance Manual.

<table>
<tr><td>Designation</td><td>ALARM (Alarm)</td></tr>
<tr><td>Output terminal</td><td>OUTPUT 11 pin</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment
Output terminal</td><td>ON</td><td>OFF</td></tr>
<tr><td>High</td><td>O</td><td></td></tr>
<tr><td>Low</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>This signal turns on if an error of level 2, 4 or 8 has occurred in the robot or robot controller.
This signal is kept ON during error detection and turns off after the error has been cleared.
For details on the errors, see the Operator's Manual.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td></td></tr>
</table>

<table>
<tr><td>Designation</td><td>SVST_A ~ SVST_B (Servo ON contact output)</td></tr>
<tr><td>Output terminal</td><td>Between OUTPUT 12 pin and OUTPUT 24 pin (contact output)</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Output terminal</td><td>Open</td><td>Short-circuit</td></tr>
<tr><td>OFF</td><td>O</td><td></td></tr>
<tr><td>ON</td><td></td><td>O</td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>While the servo power is turned on, the output terminals are short-circuited.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>The signal is of a non-voltage contact output type. The logic is a normal open contact output. The contact capacity is AC 125 V (max.), 0.5 A (max.) / DC 60 V (max.), 1 A (max.).
It takes about one (1) second from the servo ON to the time when the robot is actually ready to work. Thus, it is necessary to program a delay of 1.5 seconds or over between the SVST_A ~ SVST_B signal ON and the RUN signal ON.</td></tr>
</table>

<table>
<tr><td>Designation</td><td>EMSST_A ~ EMSST_B (Emergency stop contact output)</td></tr>
<tr><td>Output terminal</td><td>Between OUTPUT 13 pin and OUTPUT 25 pin (contact output)</td></tr>
<tr><td>Signal logic</td><td>
<table>
<tr><td>Signal judgment / Output terminal</td><td>Open</td><td>Short-circuit</td></tr>
<tr><td>OFF</td><td></td><td>O</td></tr>
<tr><td>ON</td><td>O</td><td></td></tr>
</table>
</td></tr>
<tr><td>Function</td><td>While the EMERGENCY stop pushbutton switch provided on the control panel or teach pendant is pressed ON or system input signals "Emergency stop contacts 1 & 2" are ON, the output terminals of this signal are open.
This signal can always be used, irrespective of the master mode selected by means of the MODE switch.</td></tr>
<tr><td>Signal timing</td><td> 
</td></tr>
<tr><td>Cautions</td><td>The signal is of a non-voltage contact output type. The logic is a normal close contact output. The contact capacity is AC 125 V (max.), 0.5 A (max.) / DC 60 V (max.), 1 A (max.).</td></tr>
</table>

4.3 Fabricating External I/O Signal Cable

When fabricating an external I/O signal cable, observe the following matters.

a. Connectors INPUT and OUTPUT for the TS1000 robot controller should be either the connectors attached to the controller or equivalent connectors. The equivalent of other makers (D-SUB connectors) can also be used.

Connector name	Type of connector	Maker
INPUT	XM2A–3701 (pin type connector)	OMRON
	XM2S–3711 (connector cover)	OMRON
OUTPUT	XM2A–2501 (pin type connector)	OMRON
	XM2S–2511 (connector cover)	OMRON

b. Select the cables which meet the following specifications.

- Core wires : 0.18 mm^2 ~ 0.32 mm^2 twisted wires
- Cable outer diameter : Max. 13 mm (INPUT), 11 mm (OUTPUT)
- Shield : Batch shield
- Cable length : 30 m or less

NEVER use cables which do not conform to the specifications. Otherwise, short-circuit or cable melting due to overheated cable core may be caused. Additionally, the robot may work improperly due to noise.

c. The INPUT and OUTPUT connectors are of a solder cup type which joints cables by soldering. For how to joint cables to connectors and how to shield cables, see Appendix 2.

* Cables with their single side soldered to the connector (TS1000 INPUT/OUTPUT cables) are provided optionally. (We can advise of the cable length.) Contact our after-sale service agent in your territory.

4.4 Attaching and Detaching External I/O Signal Cable

Before attaching or detaching an external I/O signal cable to or from the TS1000 robot controller, be sure to turn off the POWER switch equipped on the front panel of the controller, or the main power in the user's control panel.

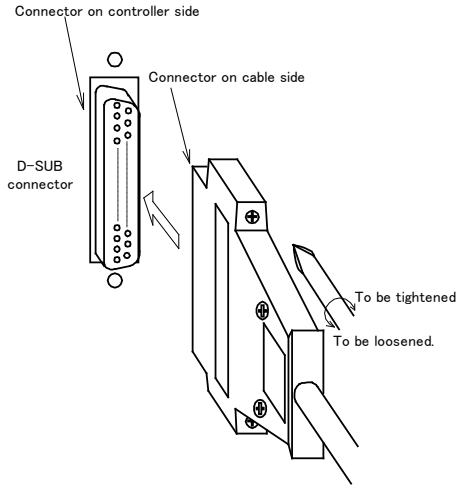


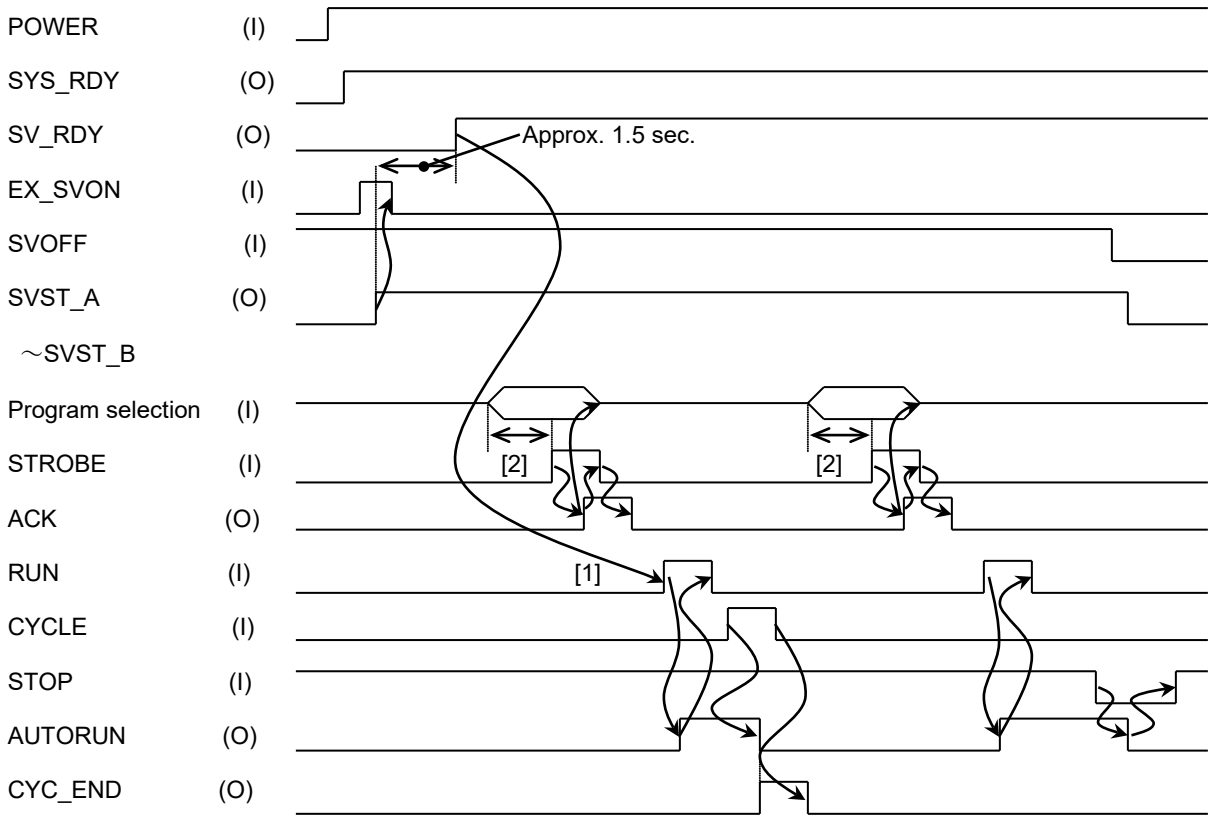
Fig. 4.3 Attaching and detaching external I/O signal cable

To connect an external I/O signal cable, completely insert the cable side connector into the controller side connector and tighten the lock screws on both sides of the connector with a screwdriver.
To disconnect the cable, loosen the lock screws, then pull out the connector. At this time, DO NOT pull the cable instead of the connector. Otherwise, an excessive force will be exerted on the cable, resulting in damage of the cable.

! **CAUTION**	When attaching or detaching the cable, be sure to hold the connector rather than the cable.

4.5 Example of Controller Operation, Using External Signals

(1) Normal operation sequence

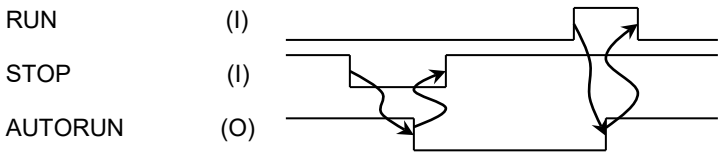


[1] To include a delay time of 1.5 seconds or over between the EX_SVON signal ON and the RUN signal ON, set ON the RUN signal only after the SV_RDY signal is ON.

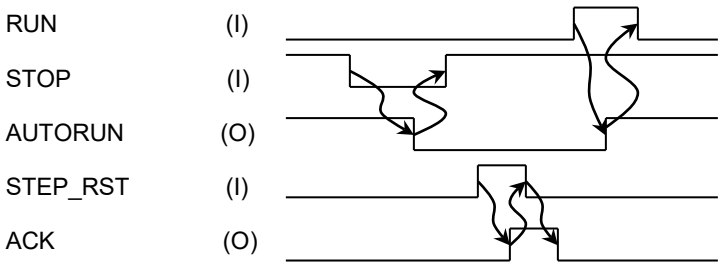
[2] Include a sufficient delay time before the program select signal starts between the program select signal input and STROBE signal input. (200 msec or so)

(2) Example of restarting the stopped robot

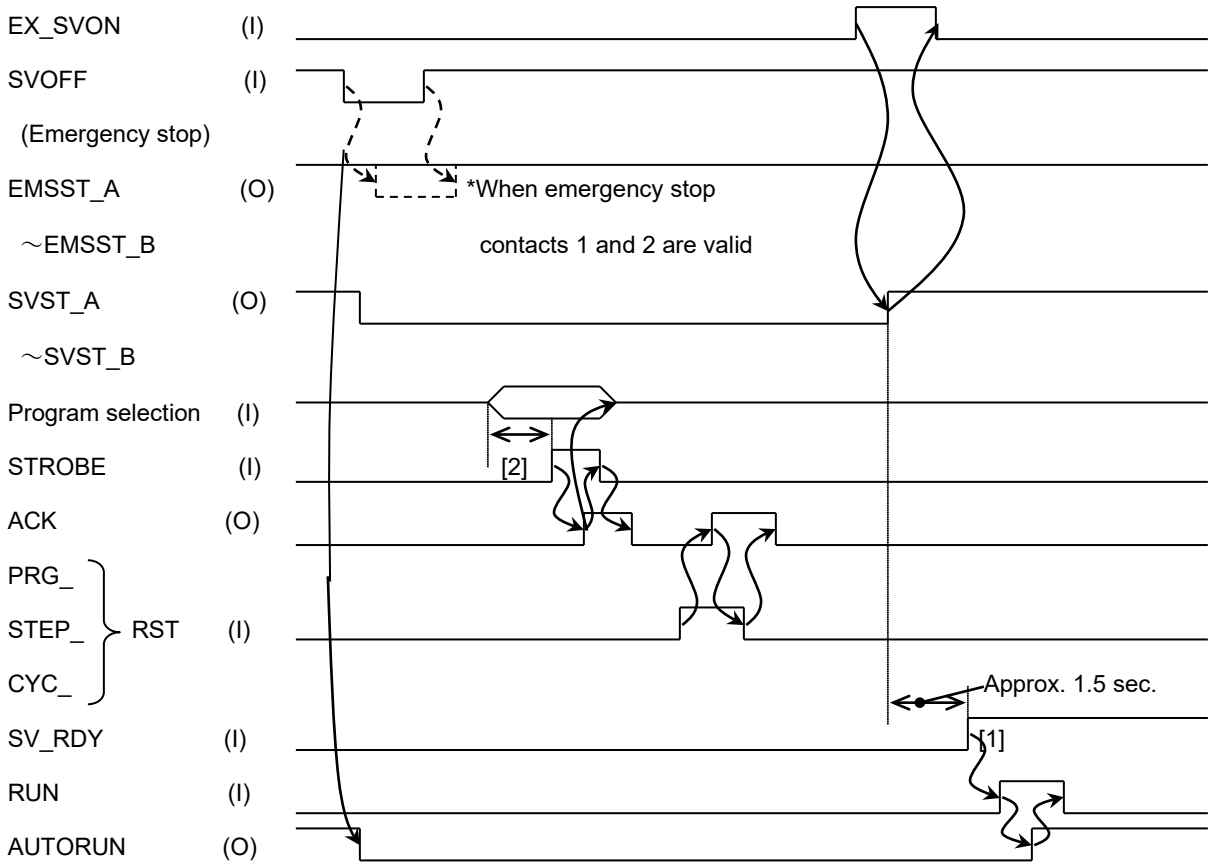
Restart after stop (to resume the operation)



Restart after stop (to start the program from the top)



(3) Example of restarting the robot after servo OFF (emergency stop)
Restart after servo OFF (emergency stop)



[1] To include a delay time of 1.5 seconds or over between the EX_SVON signal ON and the RUN signal ON, set ON the RUN signal only after the SV_RDY signal is ON.

[2] Include a sufficient delay time before the program select signal starts between the program select signal input and STROBE signal input. (200 msec or so)

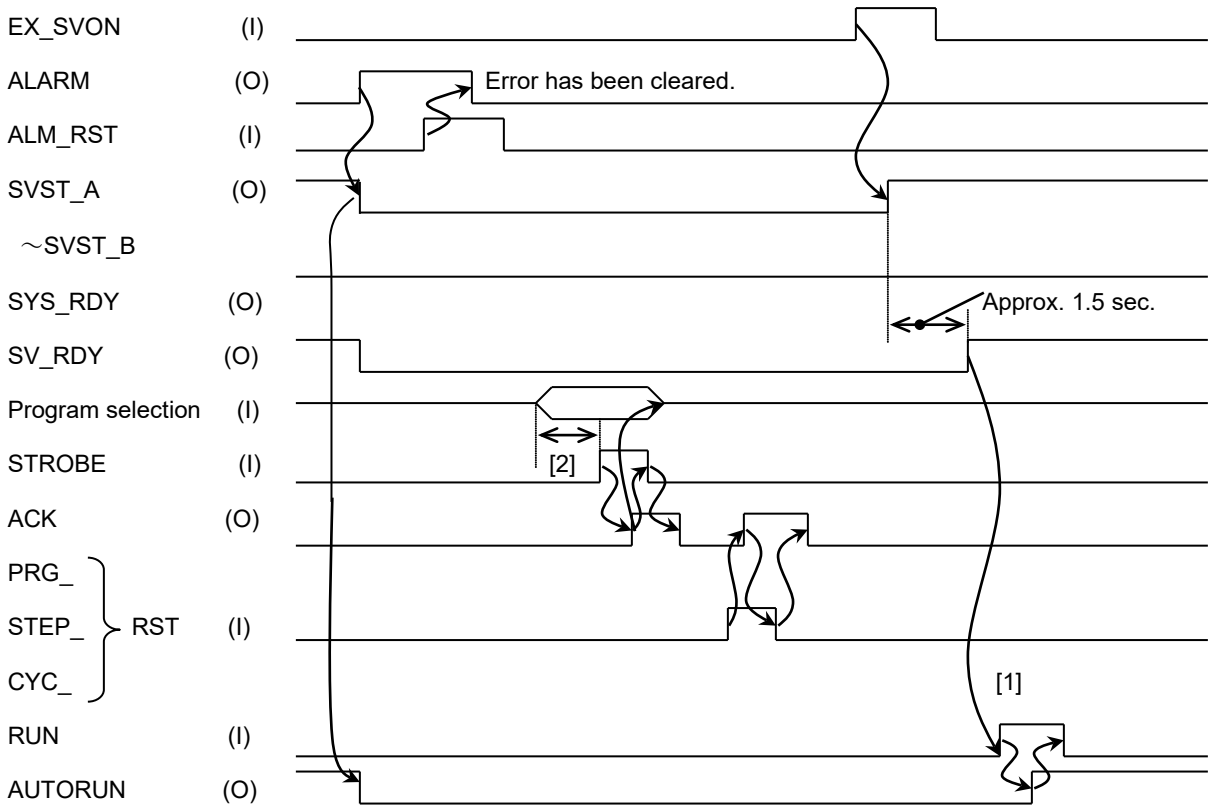
Note 1: Signals EMSST_A ~ EMSST_B (emergency stop contact output) turn on when the emergency stop contacts 1 and 2 become valid.

Note 2: Program selection, PRG_RST, STEP_RST and CYC_RST can be skipped unless necessary.

Note 3: Program selection, PRG_RST, STEP_RST and CYC_RST can be executed at any timing unless the auto operation mode is selected.

(4) Example of restarting the robot after error correction

To resume the automatic operation after an error has occurred (i.e., processing of automatic operation after error correction):



[1] To include a delay time of 1.5 seconds or over between the EX_SVON signal ON and the RUN signal ON, set ON the RUN signal only after the SV_RDY signal is ON.

[2] Include a sufficient delay time before the program select signal starts between the program select signal input and STROBE signal input. (200 msec or so)

Note 1: To continue the automatic operation, program selection, PRG_RST, STEP_RST and CYC_RST are necessary.

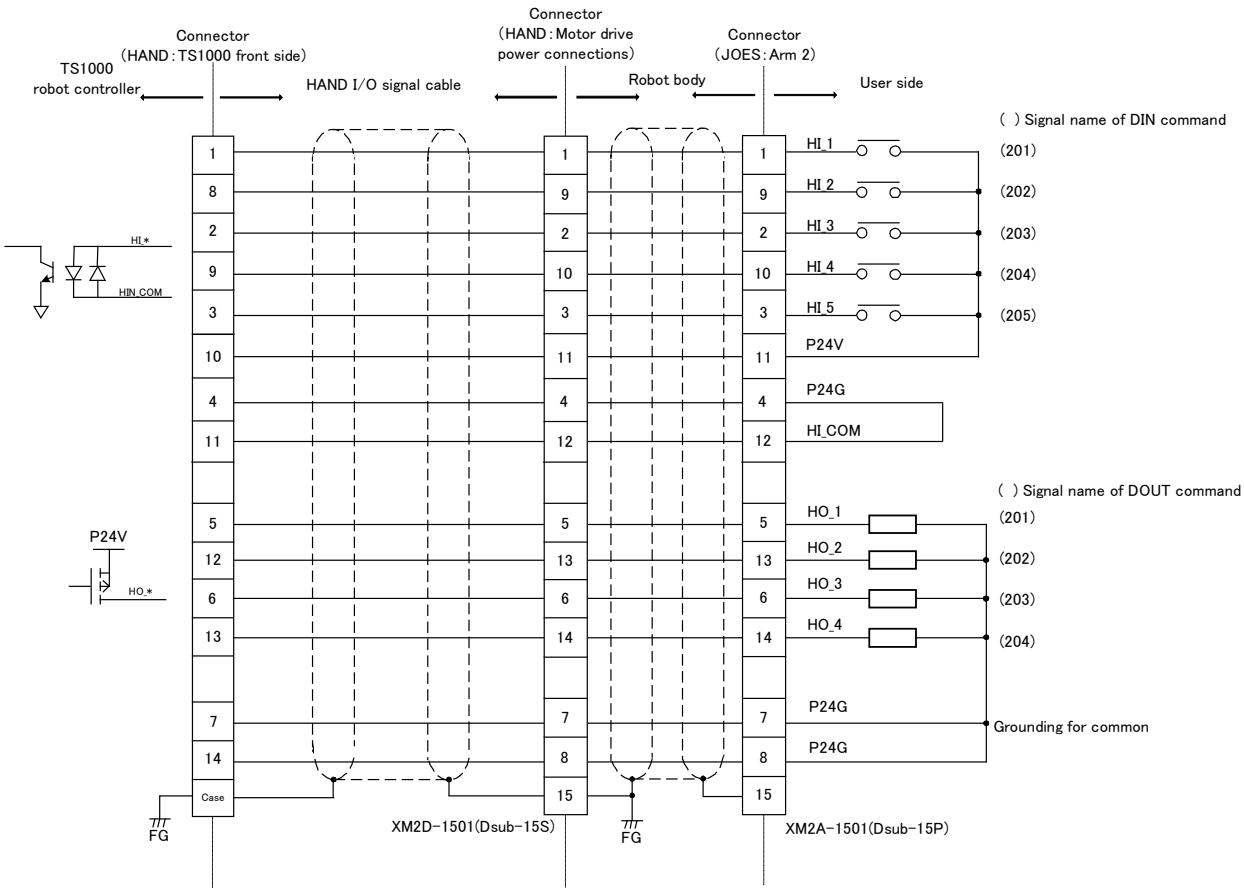
Note 2: The servo power may not be turned off with some type of error.

Note 3: Program selection, PRG_RST, STEP_RST and CYC_RST can be executed at any timing unless the auto operation mode is selected.

Section 5 Hand I/O Signal Cable

5.1 Connecting Hand I/O Signal Cable

To use the I/O signals of the TS1000 controller, exclusive cables (wiring is shown below; option) should be used. To use a user's special system, use the attached connector [XM2D–1501 (socket type connector), XM2S–1511 (connector cover)] for connecting the HAND cable (motor drive power connections) on the robot side. In either occasion, use the attached connector [XM2A–1501 (pin type connector), XM2S–1511 (connector cover)] for connecting the JOES cable at areas ahead of the arm 2.



Note: An example connection of HI_* is shown in the figure above, taking P24G as common. If P24V serves as common, connect P24V to HI_COM and P24G to HI_*.

Fig. 5.1 Connection of hand I/O signal cable

For details of the hand I/O cable connection, see the Installation and Transport Manual.

5.2 Fabricating Hand I/O Signal Cable

When fabricating a hand I/O signal cable, observe the following matters.
The descriptions below are made, however, assuming that the user uses his own special system (i.e., the hand I/O function of the TS1000 robot controller is not used). When using this function, use of exclusive cables (option) is recommended. For details, contact our after-sale service agent in your territory.

a. The HAND connector on the robot side should be either the attached connector or equivalent connectors. The equivalent of other makers (D-SUB connector) can also be used.

Connector name	Type of connector	Maker
HAND	XM2D–1501 (socket type connector)	OMRON
	XM2S–1511 (connector cover)	OMRON
JOES	XM2A–1501 (pin type connector)	OMRON
	XM2S–1511 (connector cover)	OMRON

b. Select the cables which meet the following specifications.

- Core wires : 0.18 mm^2 ~ 0.32 mm^2 twisted wires
- Cable outer diameter : Max. 10 mm (HAND, JOES)
- Shield : Batch shield
- Cable length : 30 m or less

NEVER use cables which do not conform to the specifications. Otherwise, short-circuit or cable melting due to overheated cable core may be caused. Additionally, the robot may work improperly due to noise.

c. The HAND and JOES connectors are of a solder cup type which joints cables by soldering. For how to joint cables to connectors and how to shield cables, see Appendix 2.

5.3 Attaching and Detaching Hand I/O Signal Cable

(When the hand I/S function of the controller is used)
Before attaching or detaching hand I/O signal cables to or from the TS1000 robot controller or robot, be sure to turn off the POWER switch equipped on the front panel of the controller, or the main power in the user's control panel.

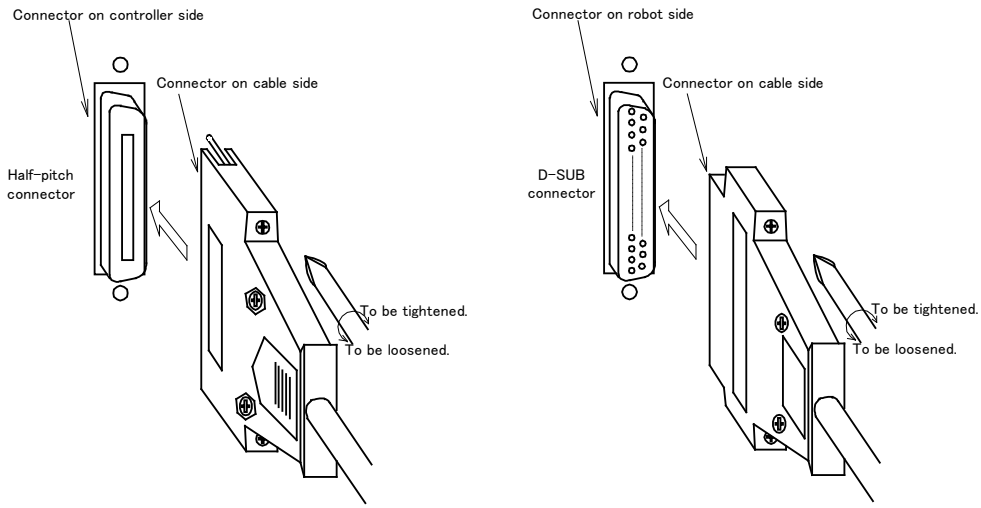


Fig. 5.2 Attaching and detaching hand I/O signal cable

To connect a hand I/O signal cable, completely insert the cable side connector into the controller side connector or robot side connector and tighten the lock screws on both sides of the connector with a screwdriver.
To disconnect the cable, loosen the lock screws, then pull out the connector. At this time, DO NOT pull the cable instead of the connector. Otherwise, an excessive force will be exerted on the cable, resulting in damage of the cable.

5.4 Hand I/O Signal

(When the hand I/O function of the controller is used)

The hand I/O signals control the end effector such as hand attached to the robot. The connector used is HAND which is located on the controller front side. The TS1000 controller is provided with five (5) input signals and four (4) output signals to control the end effector. (Wiring from the motor drive power connections of the robot to the arm 2 is already finished before shipment.)

The specifications of the input signal are the same as those of the digital input signal. (For the connection of the common wire, see Fig. 5.1.) The output type, output circuit structure and electric rating differ from those of the digital output signal, which are shown below.

All hand output signals turn off when the controller power is turned off or turned on. Therefore, the end effector should be designed so that a product will not drop from the hand at the time of power failure.

The hand I/O signals can be controlled by a program created with the robot language. In the robot language program, the hand I/O signals are specified by the signal name assigned to each signal.
For further information on the robot language, see the Robot Language Manual.

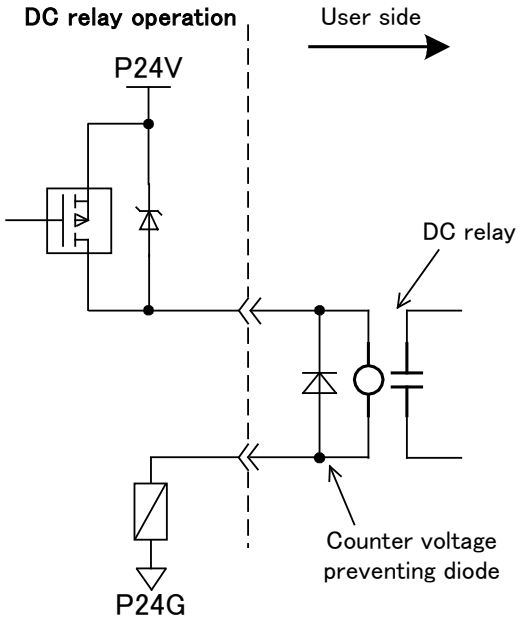
It is also possible to control a double solenoid device by combining two (2) hand output signals. When this happens, two (2) output signals bearing consecutive signal names are used for the double solenoid. Specify in the robot language program so that the two (2) output signals for the double solenoid can be exclusive during automatic operation. When the controller power is turned off and turned on, both output signals turn off.

The hand output signals can be turned on and off manually through the teach pendant. For this purpose, hand output signals to be turned on and off should be defined by the user parameter (USER. PAR) beforehand.
For the parameter setting, see the User Parameter Manual.

Note: When the double solenoid is defined by the user parameter, two (2) hand output signals are output exclusively through the teach pendant. These two (2) output signals for the double solenoid should be programmed by the robot language, however, so that they can be exclusive during automatic operation. Also, even if the double solenoid is defined, both two (2) output signals turn off when the controller power is turned on.
In the commands of OPEN1, OPEN2, CLOSE1, CLOSE2, OPENI1, OPENI2, CLOSEI1 and CLOSEI2, each set of signals (201, 202) and (203, 204) is exclusively output.
The hand output signal cannot be reset by the RESET SIG or RESET DOUT command.

The specifications of the hand output signal are as follows:

- Output type: FET output
- Electric rating: Rated voltage DC 24 V, rated current 1 A (max.)
- Output circuit structure



! **CAUTION**	If the current which exceeds the rated output current is supplied, the output device may be damaged or the printed board may be burnt. To avoid this, be sure to use within the rated output current. Amperage of four (4) hand output signals should be 1 A or less. When using the hand I/O function of the controller, NEVER use the external power supply to avoid troubles.

Section 6 Serial I/O Signal Cable

6.1 Connecting Serial I/O Signal Cables COM1, HOST and TCPRG

The TS1000 robot controller is provided with three (3) channels of RS-232C serial I/O signals COM1, HOST and TCPRG. For the connection, see the figure below. It is recommended to use commercially available cables.

Using a surge protector is also recommended in order to protect serial I/O signals from an electrostatic discharge-induced surge or switching surge.

Recommended surge protector: RSP-232-9 manufactured by Okaya Electric Industries Co., Ltd.

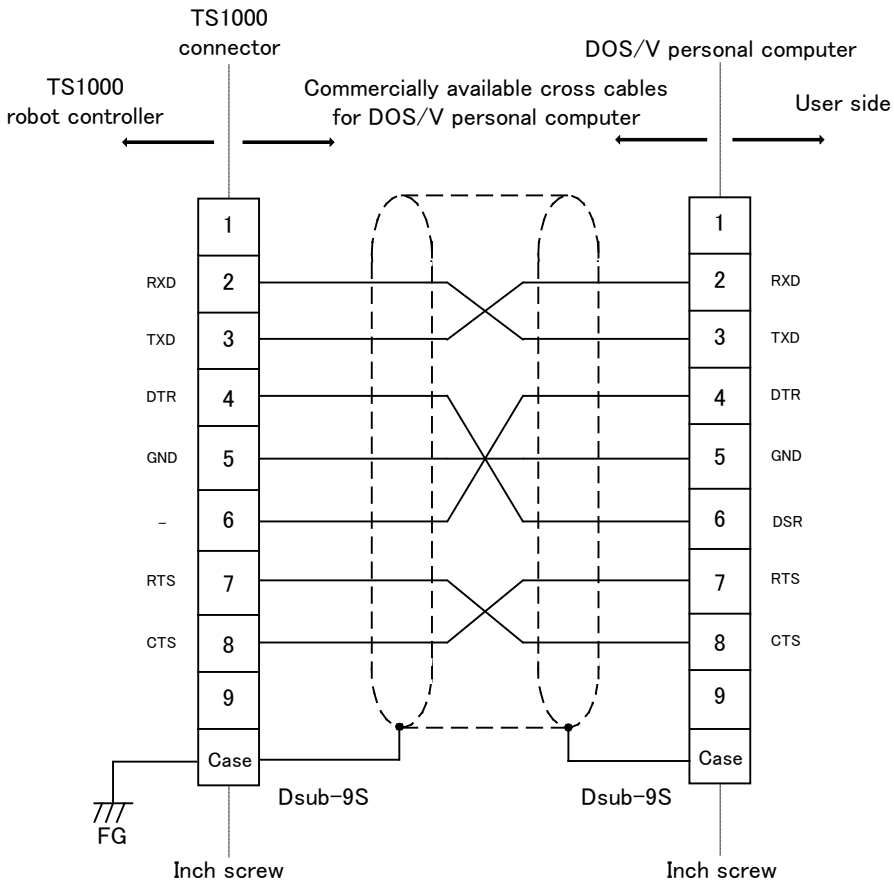


Fig. 6.1 Connection of serial I/O signals COM1, HOST, TCPRG

6.2 Attaching and Detaching Serial I/O Signal Cables COM1, HOST and TCPRG

Before attaching or detaching the serial I/O cables to or from the TS1000 robot controller, be sure to turn off the POWER switch equipped on the front panel of the controller, or the main power in the user's control panel.

The procedures for attaching and detaching serial I/O signal cables COM1, HOST and TCPRG are the same as in Para. 4.4. For details, see Para. 4.4.

! CAUTION

Be sure to turn off the main power before attaching or detaching the cables.

Attaching or detaching the cables with the main power turned on may cause damage on the substrate.

Section 7 Teach Pendant Cable

Connecting teach pendant (TP) cable

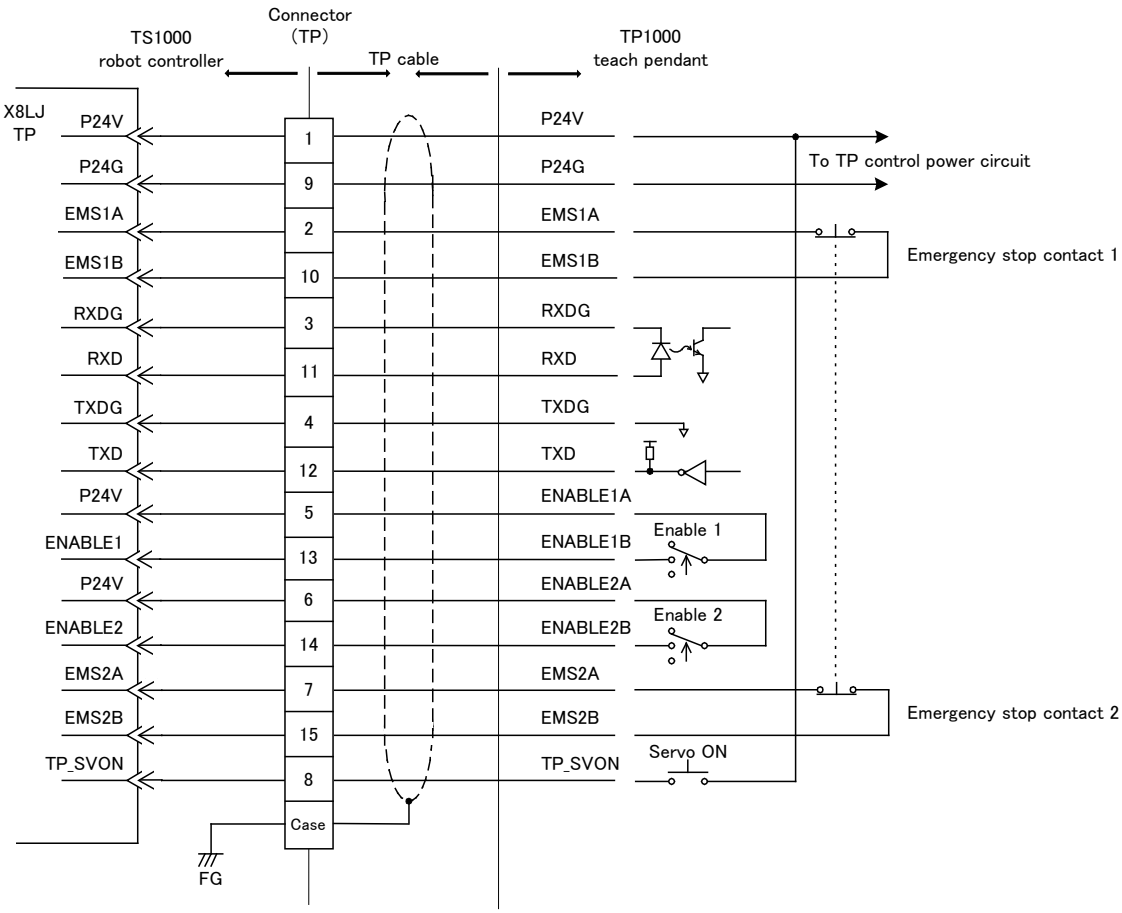


Fig. 7.1 Connection of TP1000 teach pendant I/O signals

The TP cable is a serial I/O signal cable used only for the TP1000 teach pendant (option). One side of the TP cable is secured to the teach pendant and the other side has a D-SUB 15-pin connector.

Unless the teach pendant is connected, insert the attached dummy plug to connector TP on the front panel of the controller.

When connecting the teach pendant, remove this dummy plug first. (The attaching and detaching procedures are the same as in Para. 4.4. For details, see Para. 4.4.)

The dummy plug can be removed while the POWER switch is set ON or OFF. If the dummy plug is removed while the servo power is turned on, however, the servo power is turned off automatically. (While the dummy plug or teach pendant is not connected, safety measures are effected to prevent the servo power from actuating.)

The TP disconnect switch is provided on the right side of the TP connector (when seen from the controller front side). When the dummy plug or teach pendant connector is disconnected while pressing this switch, the controller servo power remains on. Change the connector while pressing the TP disconnect switch continuously.

If the TP disconnect switch is kept pressed for long hours, however, the safety device works to turn off the servo power (i.e., emergency stop contact output is ON). The time for attaching or detaching the connector should be ninety (90) seconds or less.

! **CAUTION**	Unless the TP cable or dummy plug is connected with the TP connector of the TS1000 robot controller, the servo power cannot be turned on.

! **CAUTION**	NEVER keep pressing the TP disconnect switch intentionally.

As shown in Fig. 7.1, the teach pendant I/O signals include the I/O signals for data transmission between the teach pendant and controller, signals for identifying servo ON, emergency stop and enable input from the teach pedant, and P24V and P24G signals for supplying the power to the teach pendant.

Data transmission between the teach pendant and TS1000 robot controller is performed in the current loop system.
These I/O signals are used exclusively for the TP1000 teach pendant, and cannot be used for any other equipment.
While the teach pendant is disconnected, processing of emergency stop is effective and the servo power cannot be turned on. Unless the teach pendant is used, use the attached jumper connector (D-SUB 15 pin).

Section 8 External I/O Cable

8.1 Connecting External I/O Cable

The external I/O cable is used to connect the TR48DIOC module (option) or SHIBAURA MACHINE's designated remote I/O module (option) serving as the slave station.
To supply the controller internal power to the TR48DIOB module, the external I/O cable is connected with P24V (24 V) and P24G (0 V). (The maximum allowable current capacity of the remote I/O module is 1.2 A in total.)
When the current exceeds the maximum current capacity when the internal power supply is used, use an external power supply that can supply both 24 V and 0 V. NEVER use the internal power supply together with the external power supply. Otherwise, the power supply may be damaged.

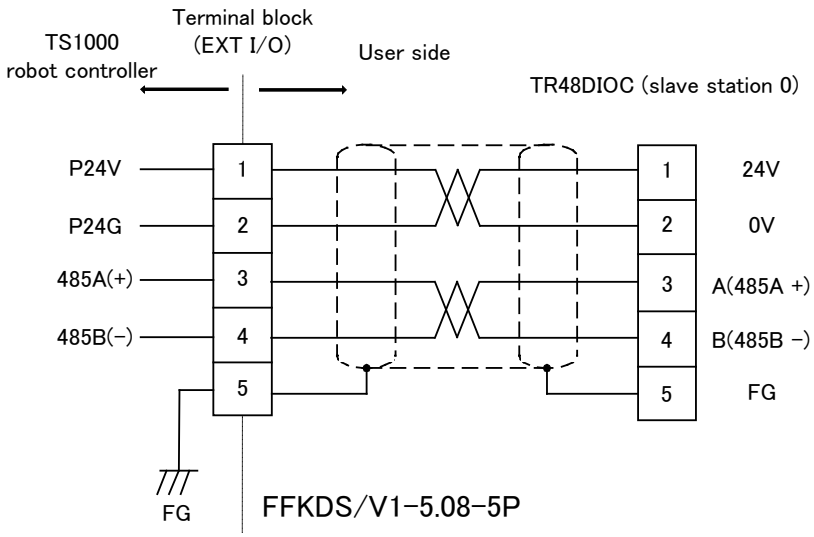


Fig. 8.1 Connection of RS485 cable

a. Use the cable attached to the TR48DIOC module, or the cable which satisfies the following specifications.

- Core wires : 0.3 mm^2 ~ 0.75 mm^2 twisted wires
- Twisted pair : Required.
- Cable outer diameter : Max. 7 mm
- Shield : Batch braided shield
- Cable length : 10 m or less in total

(When the external power is supplied to the remote I/O module and only the RS485 communication cable is extended, the total cable length can be 400 m or less.)

- Characteristic impedance : 100 Ω
- Others : For RS485 250 kbps data communication.

Recommended twisted pair cable with shield
TKVVBS 0.3 sq. × 2 P (made by Tachii Densen)

NEVER use a cable which does not conform to the specifications. Otherwise, short-circuit or cable melting due to overheated cable core may be caused. Additionally, the robot may work improperly due to noise.

b. When connecting two (2) TR48DIOC modules for I/O extension, be sure to branch from the terminal block of slave station 0, as shown in Fig. 8.2. (For the I/O extension, up to two (2) TR48DIOC modules can be connected.)

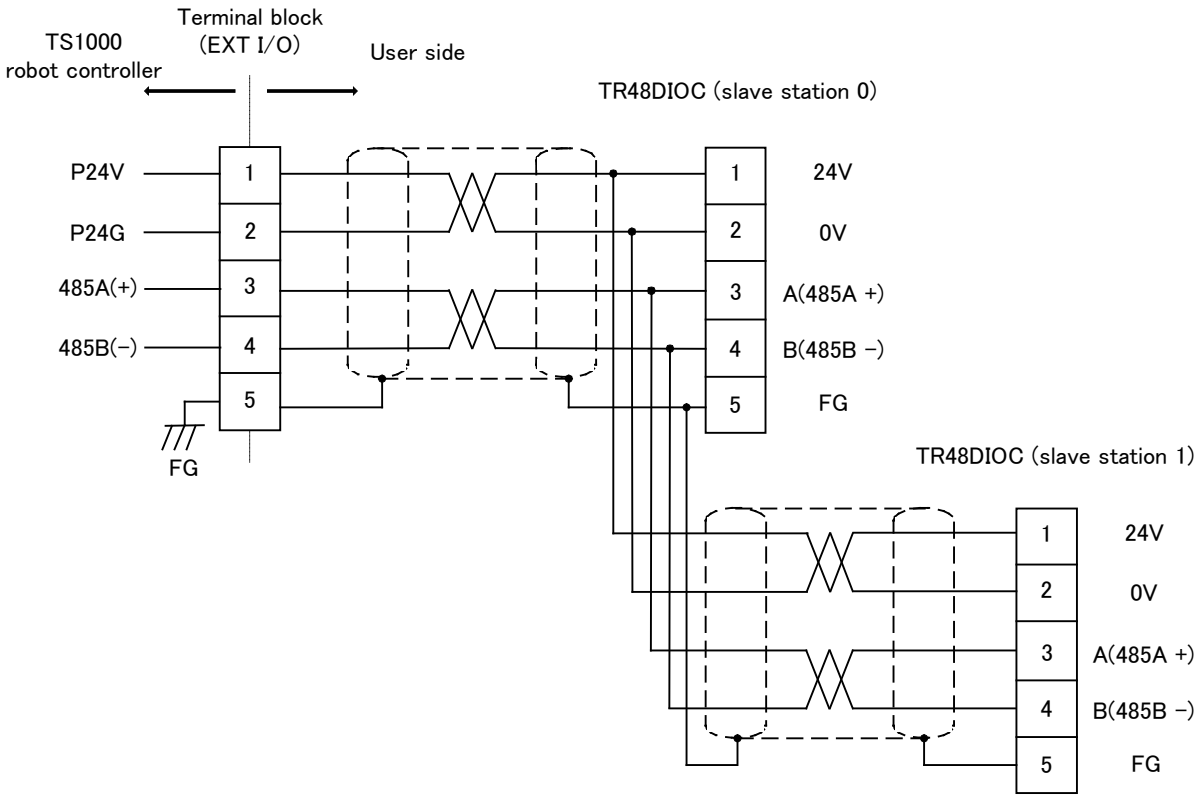


Fig. 8.2 Connection when relaying external I/O cable

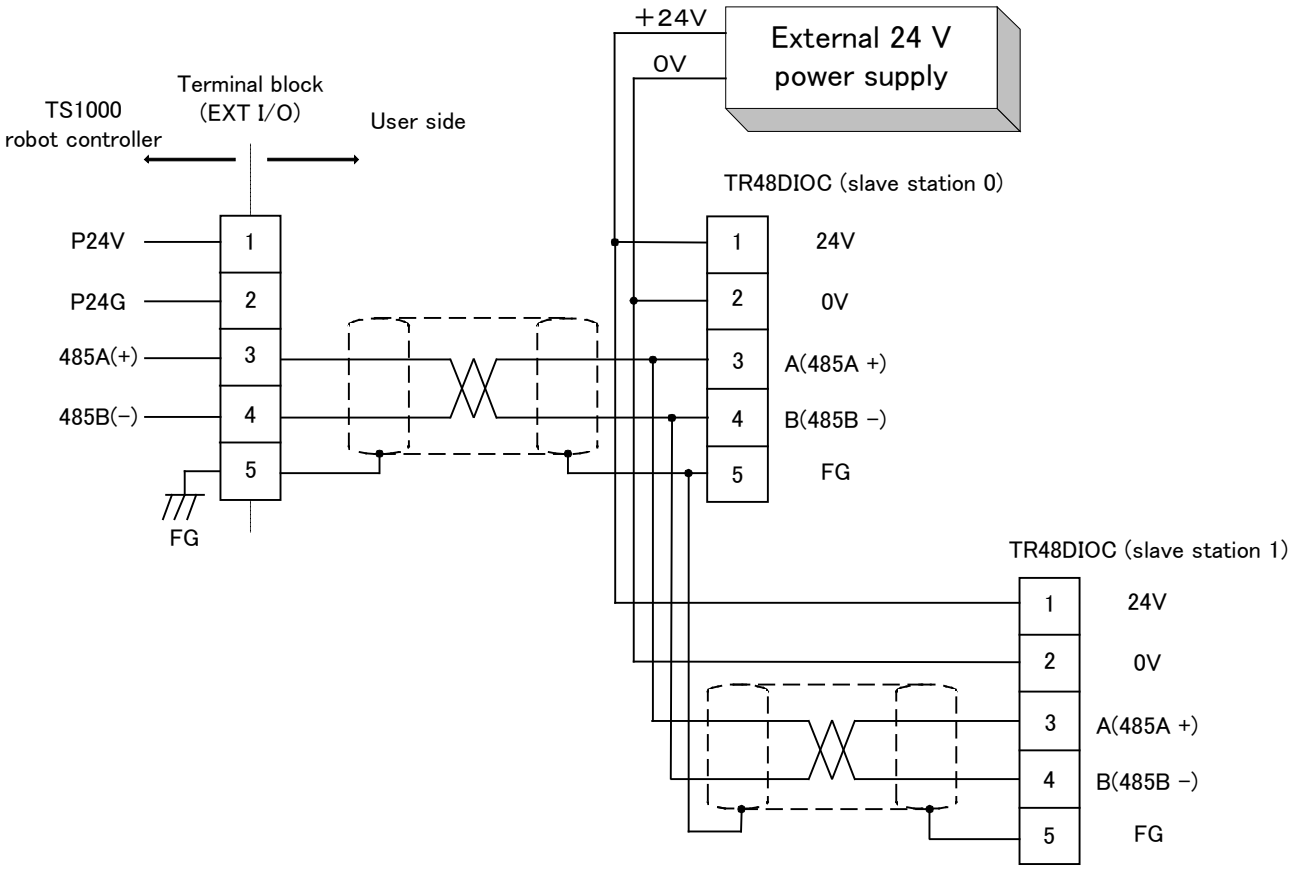


Fig. 8.3 Connection of external I/O cable when external power supply is used

8.2 External I/O Communication

If the number of digital I/Os is insufficient for the system, up to two (2) TR48DIOC modules [option; twenty-eight (28) inputs and twenty (20) outputs] can be connected as the I/O function extension unit. When the TS1000 robot controller is assigned to the master station and the TR48DIOC module to the slave station, inputs and outputs can be controlled via the RS485 serial communication.
In addition to the TR48DIOC module, a distribution I/O module designated by SHIBAURA MACHINE, which can serve as the slave station, can be used also.

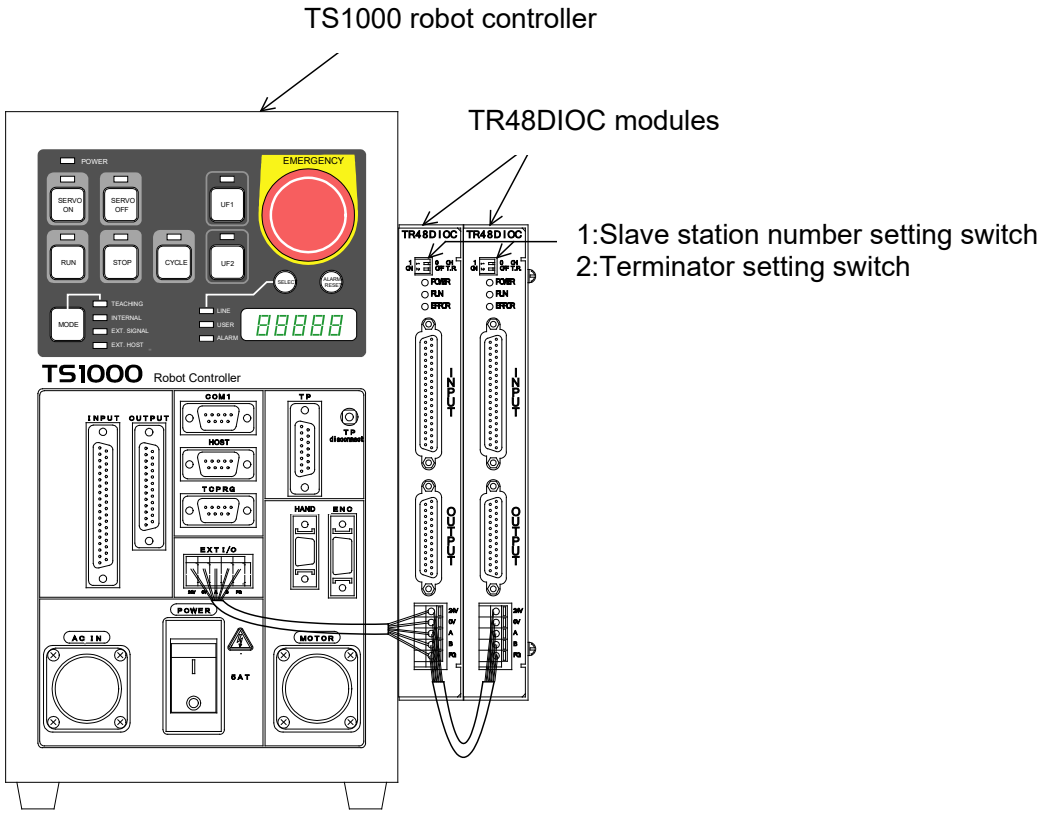


Fig. 8.4 TS1000 + TR48DIOC (2 units; when internal power supply is used)

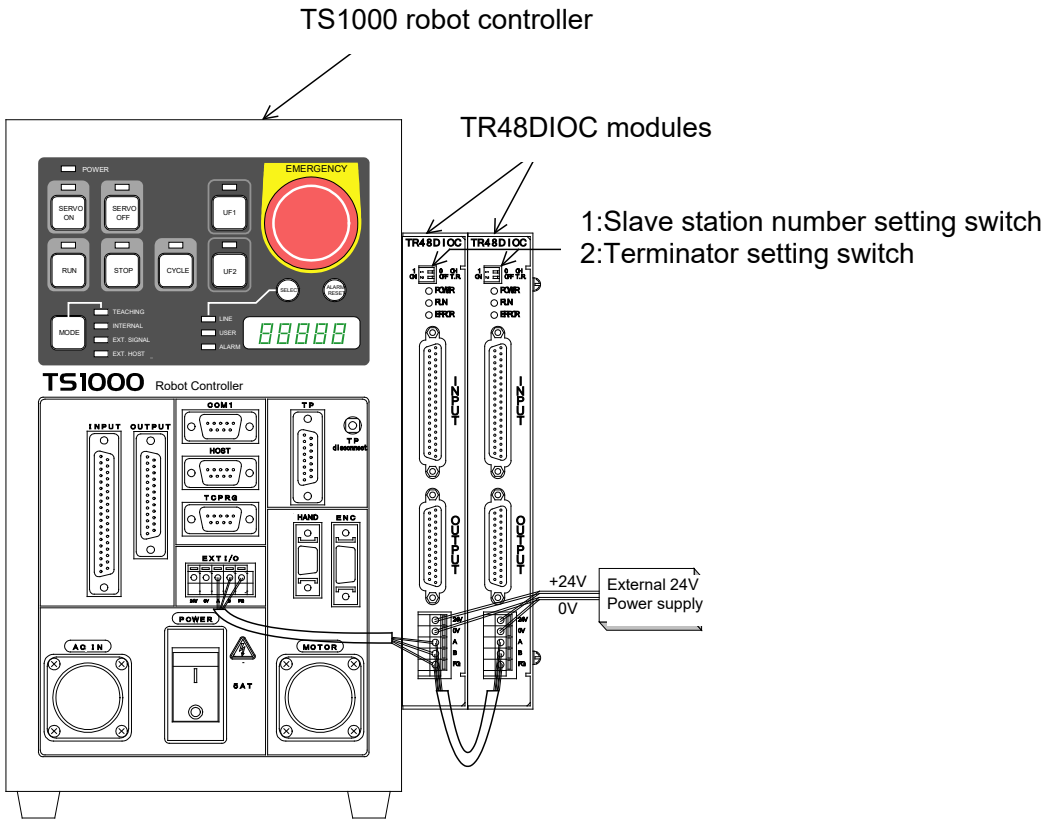


Fig. 8.5 TS1000 + TR48DIOC (2 units; when external power supply is used)

* To designate the slave station number of the TS1000, observe the following steps. Under [U12] of the USER.PAR (user parameter) file, you can find the following parameter setting related to the extension input and output.

  [U12] Extend I/O setting
  {Use/Not Use} (0: Not Use, 1: Use)
  {Not Use}
  {Not Use}
  = 0 28 20 → Corresponds to station 0.
  = 0 28 20 → Corresponds to station 1.

  If you wish to use only station 0 (i.e., one (1) TR48DIOC module), for instance, change the underlined bit corresponding to station 0 to "1".
  = 1 28 20
  = 0 28 20

  After saving the parameter file, turn off the controller power and on again. Then the above parameter becomes effective.

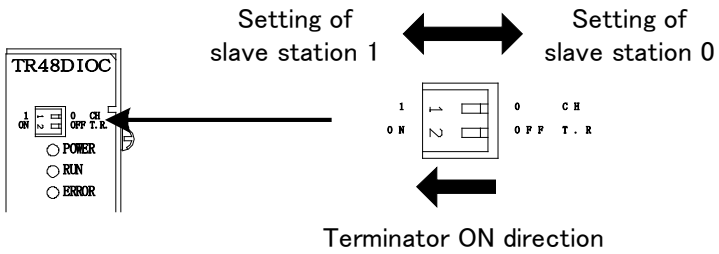
  If you wish to use both stations 0 and 1 (i.e., two (2) TR48DIOC modules), specify as follows and perform the same operation as above to make the parameter valid.
  = 1 28 20
  = 1 28 20

When using the TR48DIOC module, careful precautions should be taken on the following matters.

[1] Make sure that the slave station number set in "USER.PAR" coincides with the station number setting of the TR48DIOC module.

[2] When the TR48DIOC module or modules are connected, set ON the terminator setting switch equipped on the TR48DIOC module which is located on the extreme right side in terms of cable wiring when seen from the controller front side.

The switches (2 pins) provided on the front upper side of the TR48DIOC module represent pin 1 (upper) for slave station number setting and pin 2 (lower) for terminator setting.



The slave station setting is already described above. Specify the slave station of TR48DIOC to be used, according to the station number set in USER. PAR.

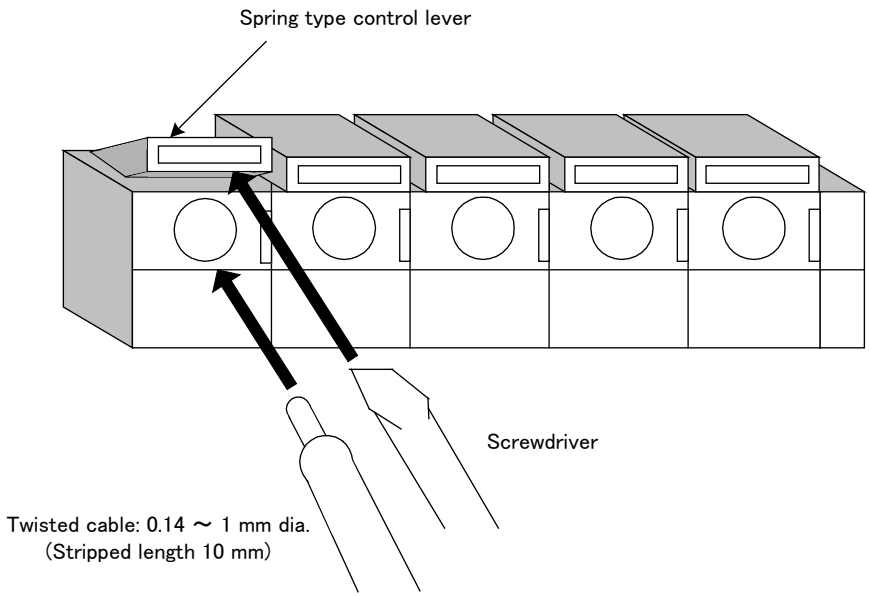
For the terminator, when only one (1) TR48DIOC module is used, set ON the terminator setting switch equipped on the module.
When two (2) TR48DIOC modules are used, see Fig. 8.4 for example. As the TR48DIOC module on the extreme right side as viewed from the controller front side is the terminal station in terms of cable wiring, set ON the terminator setting switch equipped on this module alone. The terminator of the left TR48DIOC module should remain OFF.

Be sure to execute Items [1] and [2] above, irrespective of the presence or absence of external power supply. Otherwise, the system may not function normally or go wrong.

With the above setting, make sure that both the POWER and RUN LEDs on each TR48DIOC module are illuminated.

8.3 Attaching and Detaching External I/O Cable

For the external I/O cable, use the cable attached to the TR48DIOC module, or a cable which meets the specifications.
Before attaching or detaching the external I/O cable to or from the TS1000 robot controller, be sure to turn off the POWER switch equipped on the front panel of the controller, or the main power in the user's control panel.



While holding the control lever with a screwdriver, push the twisted cable, then remove the lever from the control lever.

Fig. 8.6 Attaching and detaching external I/O cable

As shown in Fig. 8.6 above, when connecting the external I/O cable, keep holding the spring type control lever above the connector with a screwdriver and completely push the attached cable. Remove the screwdriver from the control lever then. (Lightly pull the cable and make sure that the cable is connected completely and will not come off. DO NOT pull the cable by force, however. If an unusually large force is exerted on the cable, the cable may be damaged.)
To detach the cable, hold the control lever with the screwdriver, draw out the cable and remove the screwdriver.

Section 9 Extension I/O Signal Cable

9.1 Extension Input Signal Cable

9.1.1 Connecting Extension Input Signal Cable

The extension input signal of the TR48DIOC module is called "INPUT (TR)" in this manual to discriminate it from INPUT of the TS1000 robot controller. Though it is expressed as "INPUT" on the TR48DIOC module, take utmost care not to mistake it for the external input signal of the TS1000 robot controller.
To connect INPUT (TR), use the connector attached to the TR48DIOC module [XM2A–3701 (pin type connector), XM2S–3711 (connector cover)].

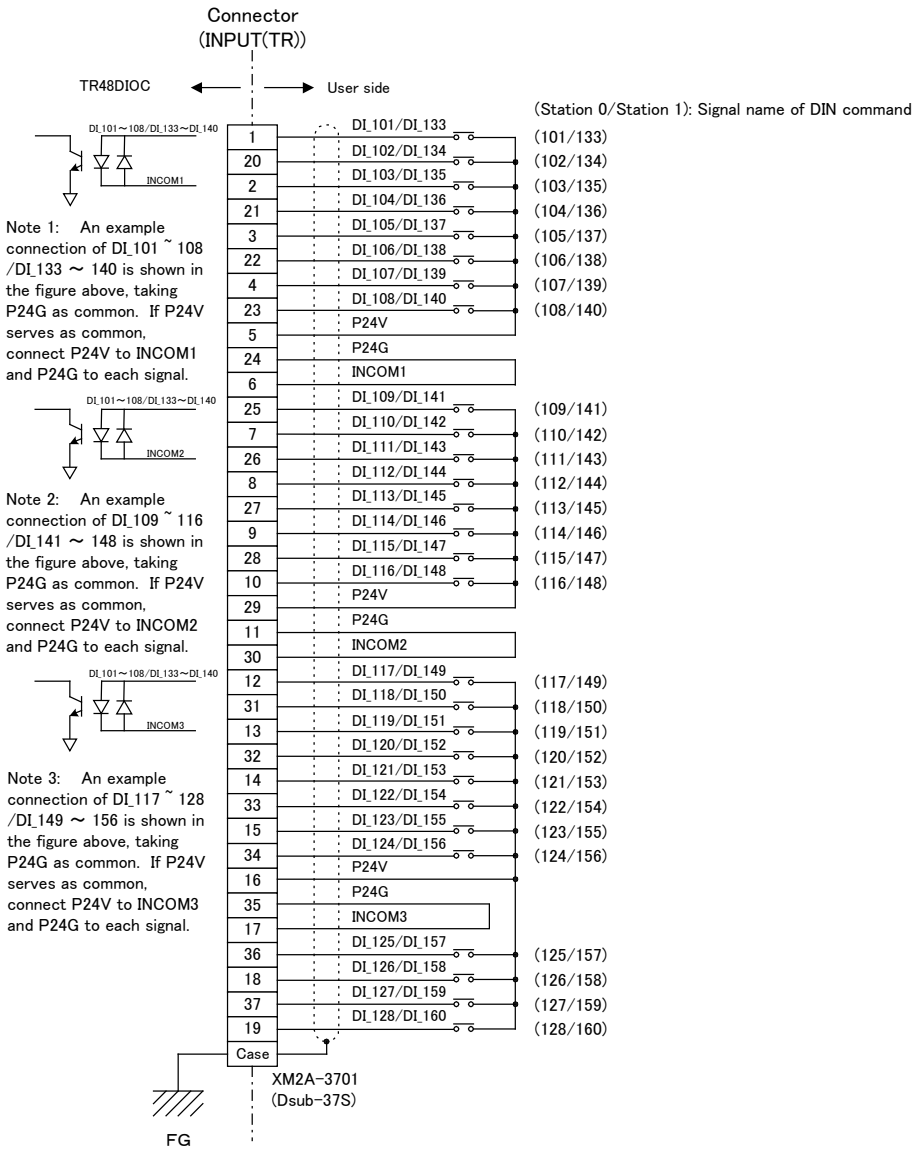


Fig. 9.1 Connection of extension input signal cable

9.1.2 Extension Input Signal

The connector used for the extension input signal is "INPUT (TR)" which is located on the front side of the TR48DIOC module.
Up to twenty-eight (28) input signals can be connected to each module. (As a maximum of two (2) TR48DIOC modules can be connected, extension of up to fifty-six (56) signals is possible.)
The specifications of the input signal are the same as those of the digital input signal in Para. 4.1.2. For further information, see Para. 4.1.2.

9.2 Extension Output Signal Cable

9.2.1 Connecting Extension Output Signal Cable

The extension output signal of the TR48DIOC module is called "OUTPUT (TR)" in this manual to discriminate it from OUTPUT of the TS1000 robot controller. Though it is expressed as "OUTPUT" on the TR48DIOC module, take utmost care not to mistake it for the external output signal of the TS1000 robot controller.
To connect OUTPUT (TR), use the connector attached to the TR48DIOC module [XM2A–2501 (pin type connector), XM2S–2511 (connector cover)].

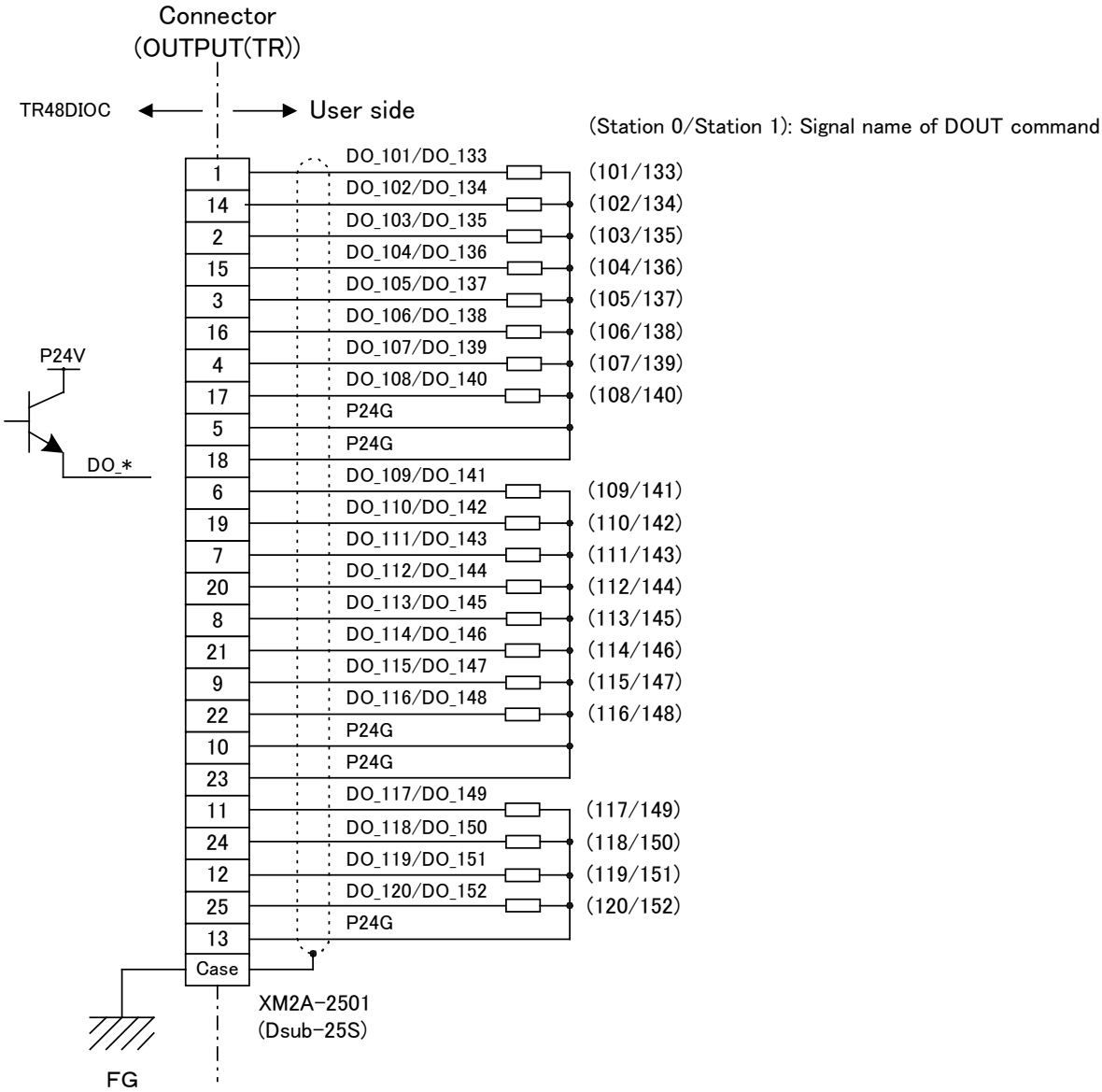


Fig. 9.2 Connection of extension output signal cable

9.2.2 Extension Output Signal

The connector used for the extension output signal is "OUTPUT (TR)" which is located on the front side of the TR48DIOC module.
Up to twenty (20) output signals can be connected to each module. (As a maximum of two (2) TR48DIOC modules can be connected, extension of up to forty (40) signals is possible.)
The specifications of the output signal are the same as those of the digital output signal in Para. 4.2.2. For further information, see Para. 4.2.2.

9.3 Fabricating Extension I/O Signal Cable

When fabricating the extension I/O signal cable, see Para. 4.3.

9.4 Attaching and Detaching Extension I/O Signal Cable

For attaching and detaching the extension I/O signal cable, see Para. 4.4.

! **CAUTION**	The maximum current capacity of the internal 24 V power supply of the TS1000 robot controller, which can be fed to the external equipment, is 3 A. Therefore, the total current used for the external I/O, hand I/O and remote I/O modules should be 3 A or less. If the current used exceeds the rated value, the internal 24 V power supply will be damaged or the fuse will be blown out. To avoid this, be sure to use within the rated capacity.

! **CAUTION**	Each of the following connectors of the TS1000 robot controller is provided with a connector cap. COM1, HOST, TCPRG, INPUT, OUTPUT Unless these connectors are used, be sure to attach the connector caps to prevent static electricity or damage.

Appendix 1 System Signal Table

Name	I/O	Function	Signal judgment
STROBE	I	Selects an execution program file from the external equipment.	ON: Short-circuit OFF: Open
PRG_RST	I	Resets a program currently stopped to step 1. The value of each variable is also reset to "0".	ON: Short-circuit OFF: Open
STEP_RST	I	Resets a program currently stopped to step 1. The value of each variable remains unchanged.	ON: Short-circuit OFF: Open
CYC_RST	I	Resets a program currently stopped to the step labeled "RCYCLE". The value of each variable remains unchanged.	ON: Short-circuit OFF: Open
DO_RST	I	Batch-resets the digital output signals (DO_1 ~ DO_16, DO_101 ~ DO_120, DO_133 ~ DO_152).	ON: Short-circuit OFF: Open
ALM_RST	I	Cancels an alarm which has occurred while the robot controller was able to work.	ON: Short-circuit OFF: Open
RUN	I	Starts a program currently stopped to execute an automatic operation.	ON: Short-circuit OFF: Open
EX_SVON	I	Turns on the servo power.	ON: Short-circuit OFF: Open
STOP	I	Stops a program during automatic operation.	ON: Open OFF: Short-circuit
CYCLE	I	Stops a program currently executed in automatic operation after one (1) cycle has finished.	ON: Open OFF: Short-circuit
LOW_SPD	I	Reduces the robot operating speed in automatic operation.	ON: Open OFF: Short-circuit
BREAK	I	Slows down and stops the ongoing robot operation.	ON: Open OFF: Short-circuit
SVOFF	I	Turns off the servo power.	ON: Open OFF: Short-circuit

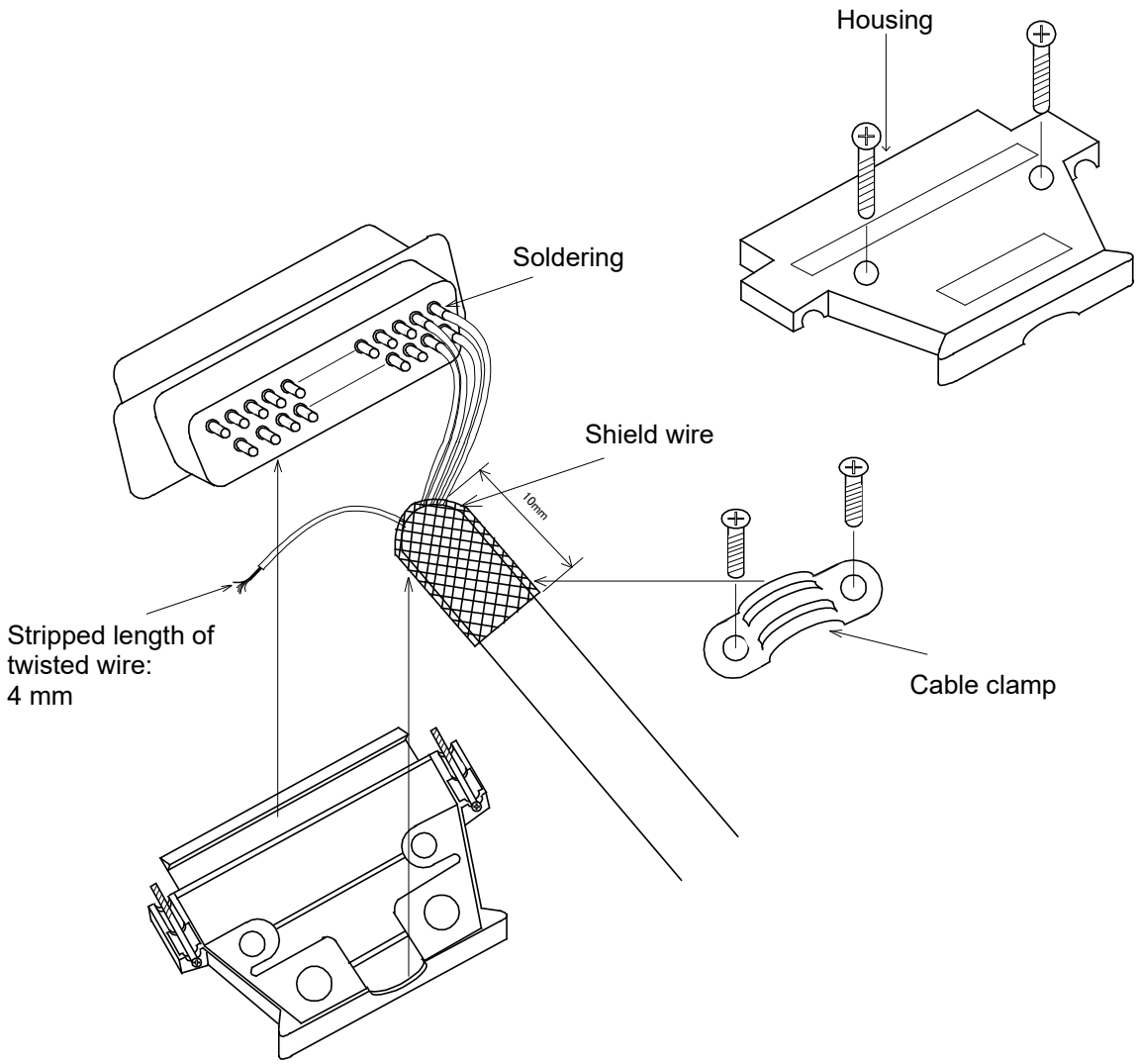
Name	I/O	Function	Signal judgment
EMS*B ~ EMS*C	I	Emergency-stops the robot.	ON: Open OFF: Short-circuit
ACK	O	Serves as a response signal to input signals STROBE, PRG_RST, STEP_RST, CYC_RST and DO_RST.	ON: High OFF: Low
SV_RDY	O	This signal is output when the servo power is turned on and the robot is ready to start.	ON: High OFF: Low
EXTSIG	O	This signal is output when the EXT. SIGNAL mode is selected by means of the MODE switch.	ON: High OFF: Low
SYS_RDY	O	System ready signal	ON: High OFF: Low
AUTORUN	O	This signal is output while the program is executed in automatic operation.	ON: High OFF: Low
CYC_END	O	This is the output signal for verifying that the program has stopped by the CYCLE signal input.	ON: High OFF: Low
LOW_ST	O	This signal is output while the robot is operating at a low speed by the input of low speed command.	ON: High OFF: Low
BT_ALM	O	Battery voltage level error output for backup power supply.	ON: High OFF: Low
ALARM	O	Controller fault output.	ON: High OFF: Low
SVST_A ~ SVST_B	O	Servo power ON contact output.	ON: Contact close OFF: Contact open Non-voltage contact
EMSST_A ~ EMSST_B	O	Used to detect the EMERGENCY stop switch equipped on the control panel or teach pendant, and system input signal of emergency stop.	ON: Contact open OFF: Contact close Non-voltage contact

I : Input signal

O : Output signal

Appendix 2 Fabricating Cable Using D-SUB Connector

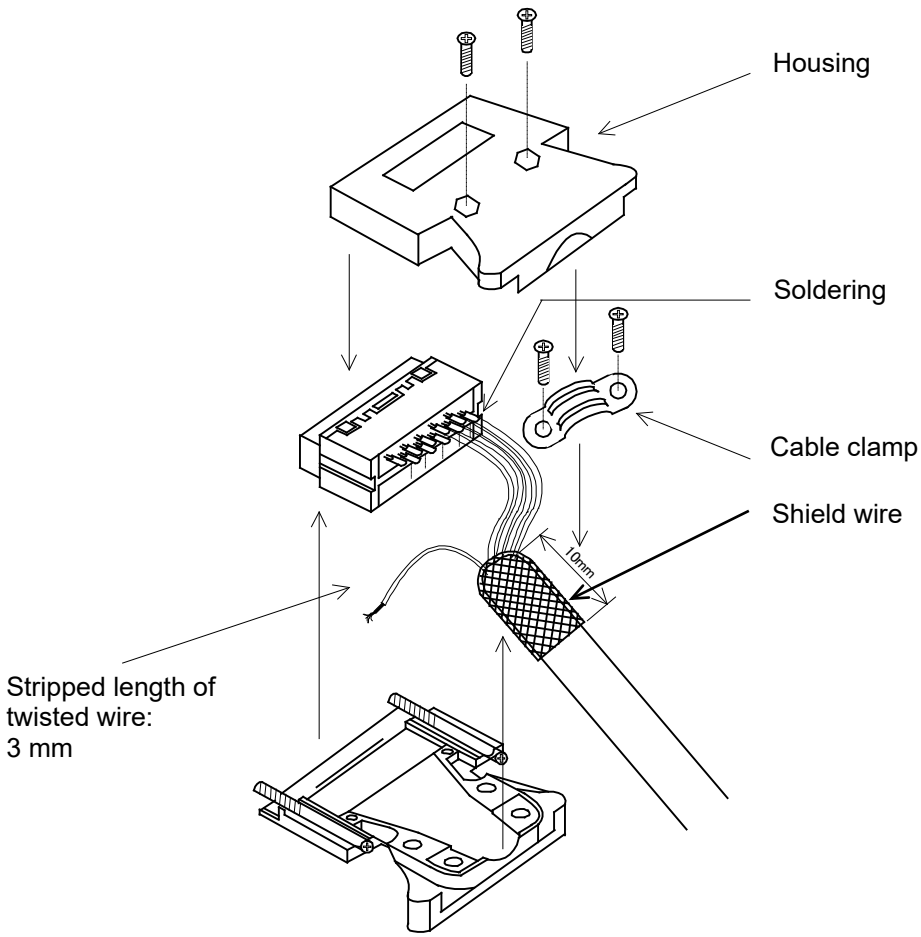
The cables to TS1000 external I/O signal connectors INPUT, OUTPUT, TR48DIOC module INPUT (TR), OUTPUT (TR), serial I/O signal connectors, hand I/O signal connector HAND (robot side motor drive power connections) and JOES connector should be fabricated as shown below.



Note: The shield wire of the cable using a D-SUB connector is connected to the ground of the TS1000 robot controller through the housing. Completely connect the shield wire to the housing using the cable clamp.
Also, it is recommended to attach a protective cover such as thermal contraction tube to the soldered part to prevent a short-circuit.

Appendix 3 Fabricating Cable Using Half-Pitch Connector

A cable to TS1000 hand I/O signal connector HAND (controller side) should be fabricated as shown below.



Note: The shield wire of the hand I/O signal cable is connected to the ground of the TS1000 robot controller through the housing. Completely connect the shield wire to the housing using the cable clamp.
Also, it is recommended to attach a protective cover such as thermal contraction tube to the soldered part to prevent a short-circuit.